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ANTI-GDF15 ANTIBODY AND A DOSAGE REGIMEN FOR THE TREATMENT OF CANCER

Abstract

The present invention relates to an anti-GDF15 antibody and to a dosage regimen for the treatment of cancer in a human patient using the anti-GDF15 antibody. The present inventors identified a mechanism by which GDF-15 blocks adhesion and transgression of predominantly T-lymphocytes into tissues. Hence, a novel treatment approach has been established by the present invention that facilitates effector T cell entry into tumor tissue upon blockage of GDF-15 using the antibody of the present invention thereby allowing the treatment of cancer in human patients.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/314,940, filed May 10, 2023, which is a continuation of International Application No. PCT/EP2021/081236, filed Nov. 10, 2021, which claims benefit of European Application No. EP 20206801.1, filed Nov. 10, 2020, European Application No. EP 21175107.8, filed May 20, 2021, and European Application No. EP 21196910.0, filed Sep. 15, 2021, each of which is incorporated by reference herein in its entirety.

SEQUENCE LISTING

[0002] The instant application contains a sequence listing which has been submitted electronically in XML ST.26 format and is hereby incorporated by reference in its entirety (said XML. ST.26 copy created on May 5, 2025, is named 217993_SL.xml and is 19,564 bytes in size).

FIELD OF INVENTION

[0003] The present invention relates to an anti-GDF15 antibody and to a dosage regimen for the treatment of cancer in human subjects with cancer using the anti-GDF15 antibody.

BACKGROUND

[0004] GDF-15, is a divergent member of the TGF-beta superfamily for which functions in appetite regulation, metabolism, cell and tissue survival, and immune tolerance have been described. GDF-15 is a homodimer, that is generated as a pro-protein, which is cleaved to a 25 kDa (2×112 aa) dimeric mature GDF-15 and 2×18 kDa (2×167 aa) pro-peptides that reside in the tissue (Tsai 2018).

[0005] To date two main categories of activity of GDF-15 have been described. The first category relates to a metabolic effect, i.e. GDF-15 mediates cachexia via changing food-intake behavior, inducing anorexia. (Johnen 2007) This effect is mediated by a brain stem specific receptor named GFRAL which was described in late 2017 (Emmerson 2017). In contrast, the second category relates to an immunomodulatory effect, i.e. GDF-15 was shown to be a mediator of immune tolerance in pregnancy (Tong et al. 2004), tissue injury (Chung et al. 2017) and inflammation (Abulizi 2016), auto-immune diseases and tumor evasion. GDF-15 inhibits leukocyte integrin activation and thereby prevents their infiltration (Kempf 2011).

[0006] In recent years increasing evidence has emerged that GDF-15 seems to play a critical immuno-regulatory role in physiologic and pathophysiologic situations and specifically in cancer. For cancer cells it would naturally be highly attractive to utilize and “hijack” such an immune-cell repellant mechanism, blocking immune-cell entry into the tumor microenvironment, and consequently preventing the immune system from removing cancer cells. In line with this, in recent years a wealth of publications has emerged indicating that high GDF-15 serum levels in various cancer types correlate with shorter overall survival and that GDF-15 is an independent factor for patient survival within various tumor types (Wischhusen et al, 2020).

[0007] Elevated GDF-15 levels are frequently reported in cancer patients. In a microarray-based study comparing 150 carcinomas from 10 anatomic sites of origin with 46 normal tissues GDF-15 showed the highest level of tumor-associated (over)expression (Welsh 2003) and several studies correlate GDF-15 serum levels and response/prognosis in cancer.

[0008] In addition, two proprietary analyses with two different academic melanoma study groups indicate that GDF-15 levels also seem to correlate with response to PD-1 antagonists.

[0009] As indicated, cancer tissues, normal organ tissues in distress and placenta are known to overexpress GDF-15, most likely in all cases to prevent an excessive immune cell infiltration to the respective tissue. Hence, the inventors considered that GDF-15 produced by above tissues does substantially reduce vascular T cell adhesion and endothelial transmigration, preventing T cell entry into the respective tissue or its immediate proximity.

[0010] Whilst an anti-GDF-15 antibody generally shows a benign and well acceptable safety profile in animal models, this mode-of-action naturally carries various potential risks when aiming at providing a suitable dosage regimen for the treatment of humans.

[0011] Rational combination partners for an anti-GDF-15 antibody will be T-cell activating compounds, such as anti-PD-1/PD-L1 checkpoint inhibitors. The efficacy of such compounds may be substantially enhanced. Yet, potentially also their toxicities may be potentiated when combining them with certain dosage regimens of an anti-GDF-15 neutralizing antibody. A second potential area of concern is the physiologic role of GDF-15 in organ protection for organs in distress. If GDF-15 is suppressed using an anti-GDF-15 antibody and organ distress occurs (e.g. myocardial infarction, infection, other significant organ damage) excessive organ infiltration by immune cells and unwelcome tissue impairment/destruction might occur.

[0012] A third potential area of concern are rare findings made in individual mouse knock-out models for GDF-15 (Wischhusen 2020).

[0013] Moreover, two important mechanisms supporting the cytotoxic effect of antibody drugs on against tumor cells are Antibody-Dependent Cell-Mediated Cytotoxicity (ADCC) and Complement Dependent Cytotoxicity (CDC). Both mechanisms can be responsible for undesirable effects on endogenous healthy cells, expressing the antibody target or binding the antibody non-specifically.

[0014] In this respect, ADCC is an important mechanism for killing target cancer cells and is based on the binding of certain antibody drugs to human FcγRIIIa receptor on immune cells (mostly natural killer cells), resulting in activation of the bound killer cell. Thereby, the recognition and binding of the target antigen on tumor cells and the linkage between target cells and immune cells by the antibody is essential for ADCC induction. The cross-linkage of target and immune cells leads to the activation of ADCC MOA (mechanism of action) pathway and finally to cell lysis.

[0015] It is additionally important to provide a stable formulation for said antibody which can be safely administered to a patient and which meets all criteria of long-term stability.

DESCRIPTION OF INVENTION

[0016] The present invention aims to overcome the unmet clinical needs to provide a safe and effective composition for use in the therapeutic treatment of human patients.

[0017] The present invention furthermore provides a safe and stable formulation for said antibody.

[0018] Based on extensive experimental tests, the inventors of the present application have found that surprisingly neither Antibody-Dependent Cell-Mediated Cytotoxicity (ADCC) nor Complement Dependent Cytotoxicity (CDC) substantially contributes to the anti-cancer effect of the anti-GDF15 antibodies of the invention.

[0019] GDF-15 blocks adhesion and transgression of T-lymphocytes into tissues. With the anti-GDF15 antibodies of the invention blocking GDF-15 a treatment approach has been established that facilitates effector T cell entry into cancer tissue, without any contributions of ADCC or CDC to the anti-cancer effect. This should increase substantially efficacy of any T cell activating agent, e.g. checkpoint inhibitors, and thus allow to provide an effective immunotherapy, either alone or in combination with checkpoint inhibitors.

[0020] At the same time, the inventor's finding that neither ADCC nor CDC contributes to the anticancer effect allows to use anti-GDF-15 antibodies which do not induce ADCC and/or CDC effectively against cancer, increasing safety overall.

[0021] Hence, the anti-GDF-15 antibodies of the invention are particularly safe and thus surprisingly combine full efficacy against cancer with safety.

[0022] Moreover, the present applicant also provides the first dosage regimen of the anti-GDF15 antibody allowing an advantageous treatment of human patients.

[0023] Accordingly, the present invention provides the following preferred embodiments:

[0024] Item 1 A composition comprising an anti-GDF-15 antibody for use in a method of treating cancer and/or cancer cachexia in a human patient.

[0025] Item 2 The composition for use according to item 1, wherein the anti-GDF-15 antibody is to be administered at a dose of 0.3, 1.0, 3.0, 10.0 or 20.0 mg/kg, preferably 3, 10 or 20 mg/kg, more preferably 10 mg/kg, and at a dosage regimen of at least one administration cycle, wherein the cycle is a period of two weeks and wherein said dose is to be administered at least once in each of the at least one cycle.

[0026] Item 3 The composition for use according to item 1, wherein the anti-GDF-15 antibody is to be administered at a dose of between 10 and 20 mg/kg, preferably 20 mg/kg, and at a dosage regimen of at least one administration cycle, wherein the cycle is a period of four weeks and wherein said dose is to be administered at least once in each of the at least one cycle.

[0027] Item 4 The composition for use according to item 1, wherein the anti-GDF-15 antibody is to be administered at a dose of between 10 and 20 mg/kg and at a dosage regimen of at least one administration cycle, wherein the cycle is a period of three weeks and wherein said dose is to be administered at least once in each of the at least one cycle.

[0028] Item 5 The composition for use according to any one of items 1 to 4, wherein said antibody does not induce antibody-dependent cell-mediated cytotoxicity (ADCC) and/or does not induce Complement Dependent Cytotoxicity (CDC).

[0029] Item 6 The composition for use according to any one of items 1 to 5, wherein said anti-GDF-15 antibody is an IgG4 isotype antibody.

[0030] Item 7 The composition for use according to item 6, wherein said antibody comprises a hinge stabilizing mutation.

[0031] Item 8 The composition for use according to item 7, wherein said hinge stabilizing mutation is a S228P mutation.

[0032] Item 9 The composition for use according to any one of items 1 to 8, wherein said anti-GDF-15 antibody comprises a heavy chain variable domain comprising a CDR1 region represented by an amino acid sequences shown in SEQ ID NO: 1, a CDR2 region represented by an amino acid sequences shown in SEQ ID NO: 2 and a CDR3 region represented by an amino acid sequences shown in SEQ ID NO: 3 and a light chain variable domain comprising a CDR1 region represented by an amino acid sequences shown in SEQ ID NO: 4, a CDR2 region represented by an amino acid sequence ser-ala-ser and a CDR3 region represented by an amino acid sequences shown in SEQ ID NO: 5.

[0033] Item 10 The composition for use according to anyone of items 1 to 9, wherein the heavy chain variable domain of said antibody comprises the amino acid sequence represented by SEQ ID NO: 6 or an amino acid sequence having at least 90% identity to the amino acid sequence shown in SEQ ID NO: 6 and the light chain variable domain of said antibody comprises the amino acid sequence represented by SEQ ID NO: 7 or an amino acid sequence having at least 90% identity to the amino acid sequence shown in SEQ ID NO: 7.

[0034] Item 11 The composition for use according to any one of items 1 to 10, wherein the heavy chain of said antibody comprises the amino acid sequence represented by SEQ ID NO: 8 or an amino acid sequence having at least 90% identity to the amino acid sequence shown in SEQ ID NO: 8 and the light chain of said antibody comprises the amino acid sequence represented by SEQ

ID NO: 9 or an amino acid sequence having at least 90% identity to the amino acid sequence shown in SEQ ID NO: 9.

[0035] Item 12 The composition for use according to any one of items 1 to 11, wherein said anti-GDF-15 antibody is obtainable by expression in CHO cells.

[0036] Item 13 The composition for use according to any one of items 1 to 12, wherein the concentration of GDF-15 in the serum/plasma of said patient is below 10 ng/mL at the end of an administration cycle.

[0037] Item 14 The composition for use according to item 13, wherein the concentration of GDF-15 in the serum of said patient is below 2 ng/mL at the end of an administration cycle.

[0038] Item 15 The composition for use according to item 13, wherein the concentration of GDF-15 in the serum of said patient is below 0.5 ng/mL at the end of an administration cycle.

[0039] Item 16 The composition for use according to any one of items 1 to 15, wherein the administration cycle is a period of three weeks and wherein said dose is to be administered at least once in each of the at least one cycle.

[0040] Item 17 The composition for use according to any one of items 1 to 15, wherein the administration cycle is a period of four weeks and wherein said dose is to be administered at least once in each of the at least one cycle.

[0041] Item 18 The composition for use according to any one of items 1 to 15, wherein said dosage regimen consists of multiple cycles and optionally of up to 52 administration cycles.

[0042] Item 19 The composition for use according to item 18, wherein said dosage regimen consists of 26 to 52 administration cycles.

[0043] Item 20 The composition for use according to item 16, wherein said dosage regimen consists of up to 34 administration cycles.

[0044] Item 21 The composition for use according to item 20, wherein said dosage regimen consists of 17 to 34 administration cycles.

[0045] Item 22 The composition for use according to item 17, wherein said dosage regimen consists of up to 26 administration cycles.

[0046] Item 23 The composition for use according to item 22 wherein said dosage regimen consists of 12 to 26 administration cycles.

[0047] Item 24 The composition for use according to any one of items 1 to 23, wherein said anti-GDF-15 antibody is to be administered in combination with a checkpoint inhibitor.

[0048] Item 25 The composition for use according to item 24, wherein said checkpoint inhibitor is selected from the groups consisting of anti-PD-1 antibody, anti-PD-L1 antibody, or anti-CD40 antibody.

[0049] Item 26 The composition for use according to items 24 or 25, wherein said checkpoint inhibitor is to be administered in the same dosage regimen as the GDF-15 antibody.

[0050] Item 27 The composition for use according to any one of items 24 to 26, wherein said checkpoint inhibitor is to be administered prior to the administration of GDF-15 antibody.

[0051] Item 28 The composition for use according to item 27, wherein said checkpoint inhibitor is to be administered within 120 min prior to the administration of GDF-15 antibody.

[0052] Item 29 The composition for use according to item 28, wherein said checkpoint inhibitor is to be administered within 30 minutes prior to the administration of GDF-15 antibody.

[0053] Item 30 The composition for use according to any one of items 1 to 29, wherein the cancer is selected from the group consisting of all brain cancers (including glioma), all cancers of the nervous system, melanoma, lung cancer, liver cancer, endometrial cancer, cervical cancer, ovarian cancer, breast cancer, renal cell carcinoma, bone and soft tissue tumors (including e.g. Ewing's sarcoma), non-small cell lung cancer and small cell lung cancer, lip and oral cavity cancer, nasopharynx cancer, larynx cancer, pharynx cancer, any type of head and neck cancer, leukemia, Hodgkin's lymphoma, Non-Hodgkin's lymphoma, bladder cancer, any type of urothelial tract cancer, cancer of the testis, thyroid cancer, kidney cancer, gallbladder and choledochus tract cancer,

multiple myeloma, oesophageal cancer, gastrointestinal tumors including stomach, colorectal, rectal and anal cancer, pancreatic cancer, prostate cancer, ovarian cancer, breast carcinoma, and carcinoma of unknown primary, and any other type of human cancer, and wherein the use is optionally also a use for the treatment of cancer-cachexia.

[0054] Item 31 The composition for use according to any one of items 1 to 30, wherein said dose of the anti-GDF-15 antibody is to be administered intravenously.

[0055] Item 32 An anti-GDF-15 antibody, wherein said antibody is an IgG4 isotype antibody having a hinge stabilizing mutation and wherein the heavy chain variable domain of said antibody comprises the amino acid sequence represented by SEQ ID NO: 6 or an amino acid sequence having at least 90% identity to the amino acid sequence shown in SEQ ID NO: 6 and the light chain variable domain of said antibody comprises the amino acid sequence represented by SEQ ID NO: 7 or an amino acid sequence having at least 90% identity to the amino acid sequence shown in SEQ ID NO: 7.

[0056] Item 33 The anti-GDF-15 antibody according to item 32, wherein the heavy chain of said antibody comprises the amino acid sequence represented by SEQ ID NO: 8 or an amino acid sequence having at least 90% identity, preferably at least 95% identity, more preferably at least 98% identity to the amino acid sequence shown in SEQ ID NO: 8 and the light chain of said antibody comprises the amino acid sequence represented by SEQ ID NO: 9 or an amino acid sequence having at least 90% identity, preferably at least 95% identity, more preferably at least 98% identity to the amino acid sequence shown in SEQ ID NO: 9.

[0057] Item 34 The anti-GDF-15 antibody according to items 32 or 33, wherein said hinge stabilizing mutation is a S228P mutation.

[0058] Item 35 The antibody according to any one of item 32 to 34, wherein said antibody does not induce antibody-dependent cell-mediated cytotoxicity (ADCC) and/or does not induce Complement Dependent Cytotoxicity (CDC).

[0059] Item 36 The antibody according to any one of items 32 to 35, wherein said antibody is obtainable by expression in CHO cells.

[0060] Item 37 The antibody according to any one of items 32 to 36, wherein said anti-GDF-15 antibody comprises a heavy chain variable domain comprising a CDR1 region represented by an amino acid sequences shown in SEQ ID NO: 1, a CDR2 region represented by an amino acid sequences shown in SEQ ID NO: 2 and a CDR3 region represented by an amino acid sequences shown in SEQ ID NO: 3 and a light chain variable domain comprising a CDR1 region represented by an amino acid sequences shown in SEQ ID NO: 4, a CDR2 region represented by an amino acid sequence ser-ala-ser and a CDR3 region represented by an amino acid sequences shown in SEQ ID NO: 5.

[0061] Item 38 A formulation comprising the anti-GDF-15-antibody as defined above or the composition for use, comprising said antibody as defined above, wherein said formulation comprises 10-50 mg/ml of the anti-GDF-15 antibody.

[0062] Item 39 The formulation according to item 38, wherein said formulation comprises histidine/histidine HCl, sucrose, arginine-HCl, and polysorbate at a pH between 5 and 6.

[0063] Item 40 The formulation according to item 38 or 39, wherein said formulation comprises 10-50 mg/ml CTL-002, 10-50 mg/ml histidine/histidine HCl, 100-200 mM sucrose, 20-80 mM arginine-HCl, and 0.01 to 0.05% w/v polysorbate 20 or polysorbate 80 at a pH between 5.0 and 6.0, preferably between pH 5.3 and 5.7.

[0064] Item 41 The formulation according to any one or more of items 38-40, which comprises 25 mg/ml CTL-002, 20 mM histidine/histidine HCl, 150 mM sucrose, 50 mM arginine-HCl, 0.02% w/v polysorbate 20, at pH 5.5.

[0065] Item 42 The formulation according to item 41, which consists of 25 mg/ml CTL-002, 20 mM histidine/histidine HCl, 150 mM sucrose, 50 mM arginine-HCl, 0.02% w/v polysorbate 20, at pH 5.5.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0066] FIG. 1: Treatment of GDF-15-producing tumors with anti-GDF-15 antibody B1-23 (murine progenitor Ab of CTL-002) substantially improves response to anti-PD1 antibodies. Anti-GDF-15 antibody B1-23 alone had only minimal effect, whereas combination with anti-PD-1 was able to substantially reverse the impairment of anti-PD-1 treatment.

[0067] FIG. 2A-FIG. 2D: Baseline GDF-15 serum level correlate with response to anti-PD1/-L1 treatment. (FIG. 2A+FIG. 2B) Study in advanced stage melanoma patients. Baseline GDF-15 level are linked to (FIG. 2A) response outcome and (FIG. 2B) overall survival (FIG. 2C+FIG. 2D) in advanced stage melanoma. Baseline GDF-15 level are linked to (FIG. 2C) response outcome and (FIG. 2D) overall survival.

[0068] FIG. 3: Affinity determination human antibody CTL-002 (=H1L5 IgG4) Binding kinetics analyzed using Biacore T3000. Different GDF-15 concentrations were applied to a flow cell with CTL-002. Association and dissociation phases were recorded to calculate the dissociation constant of the antibody. Kinetic data were evaluated by global fitting using software BIAevaluation 4.1. One representative measurement of three is shown.

[0069] FIG. 4: GDF-15 has an IC₅₀ of 7 to 14 ng/ml in the in vitro flow-adhesion assay. HUVECs are cultured over 3-days before overnight activation. T-cells are purified from healthy donors and purified on the day of experimentation. Physiological flow is generated through mounted HUVEC channel slides using a calibrated pump. Primary T-cells are pretreated with a dose titration from 0.4 to 100 ng/ml of GDF-15 for 20 mins. The HUVEC monolayer is equilibrated 20 mins with wash buffer containing 1 μ M CXCL12, then perfused 6 min with the pretreated T-cells followed by a co-culture step with wash buffer for 40 mins. Individual images are recorded every 30-secs on 1 fixed field and adhesion events are recorded as the total of number of cells per unit field. Due to technical and biological variability of the assay the IC₅₀ usually is between 7 and 14 ng/ml.

TABLE-US-00001 Assay System Experiment IC_{sub}.50 of GDF-15 Cell-based flow adhesion Study 1 (n = 3) 13.8 ng/ml Study 2 (n = 3) 7.47 ng/ml

[0070] FIG. 5: CTL-002 reverts adhesion inhibition of GDF-15 in a concentration dependent manner with an EC₅₀ of 690 to 725 ng/ml HUVECs are cultured over 3-days before overnight activation. T-cells are purified from healthy donors and purified on the day of experimentation. 50 ng/ml GDF-15 was preincubated with CTL-002 at different concentrations for 20 minutes to allow binding. Primary T-cells and HUVEC monolayer are pretreated with CTL-002/GDF-15 complex for 20 mins. The HUVEC monolayer is perfused after CTL-002/GDF-15 incubation with wash buffer containing 1 μ M CXCL12 for 5 minutes followed by a 15 minutes stasis. After washing, the pretreated T-cells are perfused for 6 minutes followed by a 50 minutes wash. The first time-point is recorded after 10 mins and then individual images are recorded every 30-secs on 1 fixed field and adhesion events are recorded as the total of number of cells per unit field for 50 minutes in total. Plots of AUC values were generated to determine EC₅₀. Three individual donors are shown. Due to technical and biological variability of the assay the IC₅₀ usually is between 690 and 725 ng/ml.

[0071] FIG. 6: CTL-002 treated animals show higher immune cell infiltration enriched in CD3+ cells.

[0072] FIG. 7: MC38-hGDF-15 show a growth advantage only in immune competent mice.

[0073] FIG. 8A and FIG. 8B: GDF-15 expression interferes with successful anti-PD-1 therapy. MC38 blank tumors respond to anti-PD-1 treatment (FIG. 8A), whereas GDF-15 secreting tumors showed severely impaired response to anti-PD-1 (FIG. 8B).

[0074] FIG. 9A and FIG. 9B: Therapy of high-GDF-15 tumors with anti-GDF-15 plus anti-PD-1 antibodies can reestablish therapy success of checkpoint inhibition. Anti-GDF-15 antibody B1-23 alone (solid line, FIG. 9A) had only minimal effect, whereas combination with anti-PD-1 (dashed

line, FIG. 9B) was able to partially reverse the impairment of anti-PD-1 treatment.

[0075] FIG. 10: Combination of B1-23 with anti-CD40/Poly(IC:LC) immunotherapy eradicated MC38 expressing human GDF-15 in contrast to anti-CD40/Poly(IC:LC) alone.

[0076] FIG. 11: Combination of B1-23 with anti-CD40/Poly(IC:LC) immunotherapy eradicated MC38 in contrast to anti-CD40/Poly(IC:LC) alone.

[0077] FIG. 12: ADCC induction in UACC-257 with test antibodies CTL-001, variants CTL-001 IgG1*PG LALA and CTL-002. Target cells were incubated with effector cells and different concentrations of test or control antibody for 6 h at 37° C. in triplicates. Afterwards, luminescence signal was measured. Here, the fold induction of the luminescence signal or the obtained relative light units (RLU) was plotted against the antibody concentration.

[0078] FIG. 13: Binding of human complement protein C1q to CTL-001, different isotype variants of CTL-001, CTL-002 and control antibody Rituximab in varying concentrations. Interaction of C1q protein and therapeutic antibody was determined in an ELISA-based approach with 10 µg/ml human C1q protein using a HRP-conjugated anti-C1q antibody for detection. Here, CTL-001 and the different isotype variants were incubated to prior coated human GDF-15 protein before C1q binding was obtained. Experiment was performed in triplicated and mean of absorption (A450 nm) was plotted against the antibody concentration.

[0079] FIG. 14A and FIG. 14B: Complement dependent cytotoxicity (CDC) induced by test antibodies and different control antibodies to GDF-15 expressing cells. Cells were incubated with the indicated concentrations of test or control antibodies in presence of 10% serum containing active baby rabbit complement proteins. Alamar Blue Viability dye was added to determine the viability of the cells. The mean fluorescence intensity (MFI) was measured after an incubation time of 6 h respectively 24 h at 37° C. and 5% CO₂. Calculated viability after 6 h [FIG. 14A] and after 24 h [FIG. 14B] of incubation are shown. Plotted is always the highest used antibody concentration.

[0080] FIG. 15: Serum concentration-time profiles of CTL-001 and CTL-002 in Cynomolgus monkeys.

[0081] FIG. 16: Observed CTL-002 PK and total GDF-15 NHP DRF study Note: the PK of CTL-002 is linear and approximately dose-proportional in the range 1-100 mg/kg. A plateau in GDF-15 capture at both 10 and 100 mg/kg dose indicates that all the available GDF-15 has been captured and increasing the dose simply increases the duration of complete target capture.

[0082] FIG. 17: Observed CTL-002 PK and total GDF-15—NHP 4 wk GLP toxicity study Observed values (symbols) for individual animals in the 4 wk GLP toxicity study. Solid lines—PK-PD model fitted to the observed data: data shown as a solid circle were included in the PK-PD model, data shown as a cross are outliers which were excluded from PK-PD modelling.

[0083] FIG. 18: Predicted human CTL-002 PK (serum concentration)

[0084] FIG. 19: Predicted human free and total GDF-15 (serum concentration) Predicted suppression of GDF-15 in the systemic circulation Three baseline levels of serum GDF-15 are considered; 0.5 ng/mL (mean level in healthy subjects and approx. 15.sup.th percentile in cancer patient cohort), 2 ng/mL (median level in cancer patient cohort; 50.sup.th percentile) and 10 ng/mL (98.sup.th percentile in cancer patient cohort)

[0085] FIG. 20: Predicted GDF-15 suppression in the tumor micro-vasculature Predicted suppression of GDF-15 in the tumor micro-vasculature Assumption: three baseline levels of systemic GDF-15; 0.5, 2 and 10 ng/mL; resulting in tumor micro-vasculature GDF-15 concentrations of 0.5, 25 and 161 ng/mL, respectively.

[0086] FIG. 21: Predicted inhibition of serum GDF-15 at planned clinical doses and for a range of baseline levels. Normal serum level of GDF-15 in healthy subjects (0.5 ng/mL) indicated as a dashed line.

[0087] FIG. 22: Sequence of the binding region of CTL-002 in various species The sequences of the binding regions of CTL-002 are shown for humans, cynomolgus monkeys, mice and rats (first

four lines from top to bottom).

[0088] FIG. 23: GDF-15 serum levels in female monkeys following the first dose of CTL-002

[0089] FIG. 24: Sequence liability map of H1L5 light chain The sequence map shows as diagrammatic representations the locations of all identified sequence liabilities in the H1L5 light chain. The locations of the domain boundaries and the CDRs as well as the type of liability are detected at a given position in relation to the overall sequence. The type of liability is indicated as follows: (I) Asn N-Linked Glycosylation, (II) Ser/Thr O-Linked Glycosylation, (III) Asn Deamidation, (IV) Asp Isomerisation/Fragmentation, (V) Pyro-Glutamate, (VI) C-Terminal Lys, (VII) Met/Trp Oxidation, (VIII) Free Thiol.

[0090] FIG. 25: Sequence liability map of H1L5 heavy chain The sequence map shows as diagrammatic representations the locations of all identified sequence liabilities in the H1L5 heavy chain. The locations of the domain boundaries and the CDRs as well as the type of liability are detected at a given position in relation to the overall sequence. The type of liability is indicated as follows: (I) Asn N-Linked Glycosylation, (II) Ser/Thr O-Linked Glycosylation, (III) Asn Deamidation, (IV) Asp Isomerisation/Fragmentation, (V) Pyro-Glutamate, (VI) C-Terminal Lys, (VII) Met/Trp Oxidation, (VIII) Free Thiol.

DETAILED DESCRIPTION OF INVENTION

[0091] Unless specifically defined herein, all technical and scientific terms used herein have the same meaning as commonly understood by a skilled artisan in the fields of gene therapy, immunology, biochemistry, genetics, and molecular biology.

[0092] All methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, with suitable methods and materials being described herein.

[0093] As used herein, each occurrence of terms such as “comprising” or “comprises” may optionally be substituted with “consisting of” or “consists of”.

[0094] The present invention provides an anti-GDF15 antibody which may be used in the treatment of cancer in human patients.

[0095] In an embodiment of the invention, the anti-GDF15 antibody does not induce antibody-dependent cell-mediated cytotoxicity (ADCC). The ADCC reporter assay to determine whether an antibody induces antibody-dependent cell-mediated cytotoxicity is not particularly limited and may be any ADCC reporter assay known in the art. An exemplary ADCC reporter assay may be the ADCC Reporter Bioassay, Core Kit (Technical Manual TM383, Promega Corporation) and is performed according to the manufacturer's protocol. In this respect, the test and control antibodies may be applied in different concentrations to the target cells and may be incubated with FcγRIIIa expressing effector cells for 6 h at 37° C. Afterwards a luciferase substrate may be added and the luminescence signal may be determined with a luminescence reader after 30 min of incubation at RT.

[0096] GDF-15 can be measured by ELISA. ELISAs which can be used to measure GDF-15 include but are not limited to the R&D systems Quantikine ELISA, an immunoradiometric assay, Luminex™ sandwich assay and electrochemiluminescence sandwich assay, as e.g. the ELECSYS® GDF15 assay (Roche Diagnostics), which was summarized by Wollert et al. (Wollert K C, Kempf T, Giannitsis E, et al. An Automated Assay for Growth Differentiation Factor 15. J Appl Lab Med An AACC Publ. 2018; 1(5):510-521. doi:10.1373/jalm.2016.022376). All mentioned assays are based on the immunosandwich principle using monoclonal or polyclonal antibodies to capture and to quantify the GDF-15. Dependent on the used reagents and their combination, free GDF-15 or total GDF-15 (free GDF-15 and GDF-15 bound to CTL-002) is measured.

[0097] In a preferred embodiment in accordance with all other embodiments of the invention, the cancer is a “solid cancer”. A “solid cancer” is a cancer which forms one or more solid tumors. Such solid cancers forming solid tumors are generally known in the art. The term “solid cancer” encompasses both a primary tumor formed by the cancer and possible secondary tumors, which are

also known as metastases. Preferred solid cancers to be treated according to the invention are selected from the group consisting of melanoma, colorectal cancer, prostate cancer, head and neck cancer, urothelial cancer, stomach cancer, pancreatic cancer, liver cancer, testis cancer, ovarian cancer, endometrial cancer, cervical cancer, brain cancer, breast cancer, gastric cancer, renal cell carcinoma, Ewing's sarcoma, non-small cell lung cancer and small cell lung cancer, carcinoma of unknown primary, preferably selected from the group consisting of melanoma, colorectal cancer, prostate cancer, head and neck cancer, urothelial cancer, stomach cancer, pancreatic cancer, liver cancer, testis cancer, ovarian cancer, endometrial cancer and cervical cancer, more preferably selected from the group consisting of melanoma, colorectal cancer, prostate cancer, head and neck cancer, urothelial cancer and stomach cancer, and most preferably selected from the group consisting of melanoma, colorectal cancer and prostate cancer.

[0098] As referred to herein, the terms “CTL-002”, “CTL-001 IgG4*”, “CTL-001 IgG4” and “H1L5 IgG4*” are used synonymously. They refer to an antibody having the heavy chain amino acid sequence of SEQ ID NO: 8 and the light chain amino acid sequence of SEQ ID NO: 9.

[0099] In a preferred embodiment in accordance with all other embodiments of the invention, the GDF-15 is human GDF-15 (also referred to herein as “hGDF-15”) and the anti-GDF-15 antibody is an anti-human GDF-15 antibody (also referred to herein as “anti-hGDF-15 antibody”).

Provision of Stable Formulation

[0100] Therapeutic proteins are complex and very heterogeneous due to post-translational modifications (PTMs) and chemical modifications. These modifications include glycosylation, deamidation, oxidation and variations of N- and C-termini. Modifications which result in relevant product-related variants are classified as critical quality attributes (CQAs) by regulators. CQAs are given narrow acceptance criteria and their variations are monitored by appropriate qualitative and quantitative methods. The provision of a stable antibody formulation is thus in many cases far from straightforward.

[0101] In a first step to approach the goal to provide a stable formulation for the inventive antibody, Applicants set out to determine which parts and sequences of the antibody were potentially at risk in the future formulation effort. To do so, an in silico determination was done. The humanised anti-GDF-15 antibody H1L5 was screened with in silico manufacturability assessment tools. The amino-acid sequence of H1L5, composed of a full-length Kappa isotype light chain and a full length IgG1 heavy chain, was screened for the sequence motifs and features of a number of potential developability issues and for aggregation risk, as set out in detail in the examples below. The Applicants found that H1L5 has a potential CDR deamidation site and an oxidation site that would have to be further evaluated. The antibody also has other potential stability issues in the form of potential oxidation and acid-labile sites as well as C-terminal clipping. It was thus clear that the present antibody would potentially not be easy to stabilize.

[0102] As evident from the initial round of experimental data several risk factors were identified which could potentially destabilize the antibody during further formulation efforts.

[0103] Thus, in a second step, the antibody H1L5 was engineered to an IgG4 backbone, as described herein elsewhere, and was then designated as CTL-002. With the IgG4 backbone three of the above identified risk factors could be eliminated, namely [0104] 1) the K448 of IgG1 has been deleted [0105] 2) the N at position 204 of IgG1 has been replaced by a D in the IgG4 antibody [0106] 3) the S at position 132 of IgG1 has been replaced by a C in the IgG4 antibody

[0107] The change from IgG1 to IgG4 thus eliminated three potential risk factors for the provision of a stable antibody formulation.

[0108] Applicants provides—on the basis of this change and the further stability studies as shown below—a stable antibody formulation. The formulation comprises preferably histidine/histidine HCl, arginine-HCl, polysorbate, and sucrose at a pH of 5-6. Further preferred formulations are described in the embodiments above and in the claims.

Examples

Drug Substance (DS)

[0109] CTL-002 is a humanized, hinge-stabilized IgG4 monoclonal antibody targeting Growth Differentiation Factor-15 (GDF-15) and relates to an antibody of the present invention.

[0110] In an exemplified liquid formulation of the CTL-002 Drug Substance, the CTL-002 antibody is presented at a concentration of about 25 mg/mL, further comprising 20 mM Histidine/Histidine HCl, 150 mM sucrose, 50 mM Arginine-HCl and 0.02% w/v Polysorbate 20, at a pH of 5.5.

Manufacturing & Control

[0111] The CTL-002 Drug Substance may be manufactured in CHO cells such as CHOK1SV GS KOTM cells. The downstream process includes 2 chromatography steps; one Protein A-based affinity chromatography (e.g. MabSelect SuRe) followed by anion exchange membrane chromatography (e.g. Sartobind Q).

[0112] Virus inactivation is achieved by e.g. Triton-X 100 treatment.

[0113] Analytical testing is performed routinely in-process and for the final release.

Stability

[0114] Drug Product stability studies are currently ongoing as follows: [0115] Pilot non-cGMP (Batch #DPS026) representative stability study for (up to) 3 years for the intended long-term storage conditions at +2-8° C., and for 12 months at +25° C. and for 6 months at +40° C. in order to provide representative stability data for the setting of provisional shelf life of Drug Product GMP batch(es). [0116] GMP (Batch #F19235) stability study for (up to) 3 years for the intended long-term storage conditions at +2-8° C., and for 12 months at +25° C. and for 6 months at +40° C. in order to confirm the stability of the proposed IMP.

Storage

[0117] CTL-002 Drug Product vials must be stored at +2-8° C. in their original secondary packaging within a secure environment, protected from light and separated from other medication or investigational product. The product should not be frozen.

Preparation & Administration

[0118] To prepare CTL-002 for intravenous administration, the CTL-002 solution is added to an infusion bag containing 0.9% NaCl. CTL-002 solution for infusion may be administered using IV bags made of polyethylene (PVC-, DEHP- and latex-free) or polyvinylchloride (latex-free) and infusion lines made of PE (PVC-, DEHP- and latex-free) or PVC (DEHP- and latex-free) material. The use of an 0.2 µm in-line filter (positive charged/uncharged PES membrane) is mandated.

[0119] Once compounded, the CTL-002 solution for infusion in infusion bags may be used immediately and administered at ambient temperature. The infusion bag might be stored up to 6 hours at room temperature and up to 24 hours at +2-8° C. but should be used no longer than 24 hours after preparation.

Nonclinical Pharmacology

[0120] The present inventors identified a mechanism by which GDF-15 blocks adhesion and transgression of predominantly CD8⁺-T-lymphocytes into tissues. With CTL-002 blocking GDF-15 a novel treatment approach has been established that facilitates effector T cell entry into tumor tissue. This may substantially enhance the efficacy of any T cell activating agent, e.g. checkpoint inhibitors.

[0121] Specifically, in a flow-adhesion assay, different immune cell subsets pre-treated +/- GDF-15 were perfused over an activated layer of endothelial cells or recombinant adhesion molecules. Adhesion and transmigration processes were monitored by live imaging microscopy. Adhesion of T cells to the endothelial cell layer was significantly impaired by addition of GDF-15. Among T-cell subsets CD8⁺ T-cells were most affected while adhesion of other immune cells was not reduced. Inhibitory effects of GDF-15 on CD8⁺ T-cell adhesion were comparable to potent blockade of LFA-1 by TS1/18 antibody and could be rescued by the anti-GDF-15 antibody CTL-002 with an EC50 of .sup.~700 ng/ml.

[0122] This initial finding has been further substantiated with data from relevant animal models, in which the anti-GDF-15 antibody CTL-002 or a mouse surrogate induced strong increase of tumor infiltrating lymphocyte numbers and increased the response to T cell activating therapies. Neutralization of GDF-15 in HV18-MK melanoma-bearing humanized mice by CTL-002 resulted in a strong increase of tumor infiltrating leukocyte numbers. Subset analysis revealed an over proportional enrichment of T-cells, especially CD8+–T cells (see FIG. 6).

[0123] Further, syngeneic mouse tumor models with genetically modified mouse colon tumor MC38 cells expressing human GDF-15 (MC38-GDF15), showed an increased response to otherwise diminished responses towards anti-PD-1 or anti-CD40/poly(IC:LC) combination therapy. No adverse effects were observed in any of the animals (FIG. 1, 13, 14).

[0124] Hence, CTL-002 is developed to neutralize the pathological effects mediated by GDF-15. The biological activity of GDF-15 adhesion and transmigration processes were monitored by live cell imaging microscopy in an in vitro flow adhesion system with primary immune cells and key parameter of the inhibition of GDF-15 effects by CTL-002 were determined in this system.

[0125] Moreover, secondary pharmacology studies examined the ability of CTL-002 to elicit CDC and ADCC, on-target/off-tissue binding as well as off-target binding. Safety pharmacology assessments were included in standard repeat-dose toxicity studies.

[0126] Overall, these studies provide a thorough characterization of the mechanism of action of CTL-002 as well as a well-supported rationale for its clinical examination in patients with cancer and GDF-15 elevation in the tumor microenvironment.

Binding of the Drug CTL-002 to the Target GDF-15

[0127] As shown in Table 1, GDF-15 and CTL-002 form a main complex of two CTL-002 antibodies and two dimeric GDF-15 molecules in solution. Other complexes seem to be less favorable and only one additional complex of three CTL-002 and three GDF-15 was reliably detected. This complex was maximal during equimolar incubation of CTL-002 and GDF-15 but was still below 8% and decreased when the ratio was changed in either direction. Above three molar excess of the antibody, all GDF-15 is complexed by CTL-002 molecules and no tendency for formation of high molecular weight aggregates was observed.

TABLE-US-00002 TABLE 1 Overview of formed complexes of CTL-002 and GDF-15 Simplified Visualization																		
   																		
 Estim. MW 506 kDa 337 kDa 169/193 kDa 144 kDa 24.6 kDa Ratio [CTL-002].sub.3- [CTL-002].sub.2- [CTL-002].sub.1- CTL-002 GDF-15 CTL002/GDF- [GDF-15].sub.3 [GDF-15].sub.2 [GDF-15].sub.1/2 15 1:30 1.1% 15.7% 4.2% 0% 79.1% 1:3 4.9% 70.3% 4.3%																		
0% 20.6% 1:1 8.0% 74.3% 4.0% 13.0% 0.6% 3:1 2.9% 27.5% 0% 69.6% 0% 30:1																		
0.4% 2.9% 0% 96.7% 0%																		

[0128] Affinity of CTL-002 to recombinant human GDF-15 was determined by measuring surface plasmon resonance on a Biacore T3000. In addition, the affinity to cynomolgus rat and mouse GDF-15 was measured (see Table 2).

[0129] All experiments were performed in 10 mM HEPES buffer, pH 7.4, 150 mM NaCl, 3.4 mM EDTA and 0.05% Tween. anti-human IgG Fc specific antibody (Jackson, Order #109-005-008, Lot #111148) was covalently immobilized by EDC/NHS chemistry on Biacore CM5 Chips (GE Healthcare, Order #61144275, Lot #10236645). For kinetic characterization of the antigen-antibody interaction pulses of increasing GDF-15 concentrations (e.g. 156.3 pM, 312 pM, 625 pM, 1,250 pM) were injected at a flow rate of 30 µl/min. After each measurement cycle (8 min of association followed by 30 min of dissociation) the antibody-antigen complex was resolved by regeneration of the surface with 10 mM glycine-HCl at pH 2.0. For calculation of the dissociation constant of CTL-002 the association and dissociation phases were recorded and evaluated by global fitting using the software BIAevaluation 4.1. For global fit analysis only, these antigen concentrations were taken into account, which allowed the analysis following the Langmuir 1:1 binding model or 1:1 binding with drifting baseline.

[0130] K.sub.D values for human GDF-15 are shown in Table 2.

TABLE-US-00003 TABLE 2 Overview of species affinities of CTL-002 CTL-002 K.sub.D (human) 38.3 pM K.sub.D (Cynomolgus) 107.7 pM K.sub.D (rat) 449 pM K.sub.D (mouse) 9.76 nM

[0131] An overview and comparison of the three antibodies used in a panel of non-clinical studies is given in Table 3.

TABLE-US-00004 TABLE 3 Overview on antibody details used in in vitro and in vivo experiments B1-23 CTL-001 CTL-002 Binding domain Discontinuous epitope (EVQVTMCIGACPSQFR ---38 amino acids--- TDTGVSLQTYDDLAKDCHCI) Species mouse humanized humanized Isotype IgG2a (rec IgG1) IgG1 IgG4S228P Alternative H1L5, B1-23-H1L5 CTL-001 IgG4*, CTL-001 names IgG4, H1L5 IgG4* Characterization Epitope mapping by studies proteolytic excision ADCC, C1q and CDC ADCC, C1q binding ADCC, C1q binding and and CDC CDC Immunoblot and Immunoblot and ELISA of human GDF-15 ELISA of human GDF- ELISA of human and 15 mouse GDF-15 Flow cytometry stain Flow cytometry stain of surface associated of surface associated GDF-15 GDF-15 Immunoblot and ELISA of different TGF-beta proteins Inhibition of LFA-1 activation by D-Storm microscopy Inhibition of adhesion of flowing T cells to preactivated HUVECs In vivo combination with anti-PD-1 to treat transgenic MC38 secreting hGDF-15 in mice In vivo combination with anti-CD40/Poly(IC:LC) to treat transgenic MC38 tumors secreting hGDF-15 in mice In vivo neutralization In vivo neutralization of cachexia in mice of cachexia in mice In vivo preliminary PK In vivo preliminary PK study in cynomolgus study in cynomolgus monkeys monkeys In vivo treatment with CTL-002 of humanized mouse human melanoma model Non-GLP single-dose dose-finding study in Cynomolgus monkeys PK and Dose-range finding in rats Mouse PK 4-Week Repeat-Dose Toxicity Study of CTL-002 in Cynomolgus Monkeys

[0132] In summary, CTL-002 is a humanized IgG4 antibody derived from mouse antibody B1-23 with high specificity for human GDF-15. CTL-002 binds with picomolar affinity to human, cynomolgus and rat GDF-15 (38.3, 108 and 449 pM, respectively), and with low nanomolar affinity to mouse GDF-15 (9.76 nM). CTL-002 binds to a discontinuous conformational epitope at the carboxy terminus of the mature GDF-15 and specifically recognizes the physiological dimeric conformation.

In Vitro Biological Activity of CTL-002 on GDF-15 Mediated Inhibition of T Cell Adhesion

[0133] Two proprietary analyses indicated in addition that GDF-15 levels also seem to correlate with non-response to PD-1 antagonists. This again is in line with the concept that GDF-15 acts as immune- and T cell repellent and keeps CD8+/CD4+ T cells out of the tumor, preventing PD-1 antagonism-based activity (FIG. 2).

[0134] A flow adhesion assay system was used to mimic the dynamics at the blood vessel wall separating the immune cells from the tumor tissue and to analyze the effect of neutralization of GDF-15 by CTL-002. To evaluate the potency of GDF-15 adhesion inhibition and the sensitivity of the system, the IC₅₀ of GDF-15 in the assay system was determined to 13.8±2.3 ng/ml (FIG. 4).

[0135] To evaluate the potency of CTL-002 in preventing GDF-15 mediated inhibition of T cell adhesion, the EC₅₀ of CTL-002 was determined in the flow-adhesion assay with T cells. As a result, in average a concentration of 707±17 ng/ml of CTL-002 was effective to increase the T cell adhesion by 50%, as was shown in different donors (see FIG. 5).

In Vivo Pharmacology

[0136] Among all TGF-beta superfamily members orthologous GDF-15 molecules show the lowest sequence conservation. While mature rat, mouse and human TGF-beta1 and BMP-2 proteins are 99-100% sequence identical between humans and mice, homology is below 70% for GDF-15 (Böttner 1999). Biological differences between different species thus cannot be ruled out. Murine

GDF-15 shows rather low sequence identity of 67.9% with its human counterpart reflected also in the fact that the anti-GDF-15 antibodies show less affinity for the mouse homologue. Two different *in vivo* approaches were followed to investigate pharmacodynamic effects.

[0137] First, a humanized mouse model was used, where immunodeficient mice are engrafted with human cord-blood derived CD34+–hematopoietic stem cells.

[0138] Three months after reconstitution these mice developed a functional human-like immune system containing all major human immune cell subsets (Wang 2018). These mice were then inoculated with a patient-derived human melanoma cell line, HV-18MK, that has been shown to secrete high levels of GDF-15, was inoculated. Mice were treated beginning three days later with isotype control or CTL-002 two times a week and after four weeks tumors were harvested and analyzed for immune cell infiltration by flow-cytometry.

[0139] Although the tumor size was unaffected, the CTL-002 treated groups showed a 9-fold increase in human tumor infiltrating CD45+ cells (FIG. 6). Within the infiltrating cell population, T-cells were enriched 4-fold. In a follow-up study, analysing the infiltrating immune cell population in more detail, the increase in infiltration of CD45+ and the enrichment in CD3+ was confirmed, albeit less pronounced with 3-fold increase of CD45+ and 4-fold enrichment of CD3+ T cells.

[0140] As a second experimental model, a genetically modified mouse cell line overexpressing human GDF-15 was generated for testing the therapeutic efficacy of anti-GDF-15 antibodies.

[0141] In this respect, MC38 colon adenocarcinoma cells are the preferred mouse tumor cells to analyze immune checkpoint blocker activity (Selby 2016) and were used to generate *in vivo* data to support the development of the approved anti-PD-1 antibody. Overexpression was implemented by stable transfection and did not affect *in vitro* proliferation compared to control treated MC38.

[0142] Age matched immunocompetent C57BL/6 and immunodeficient NCI nu/nu mice were inoculated subcutaneously with a suspension of 5×10^5 MC38 blank colon tumor cells or the transgenic derivative MC38.sup.GDF-15 expressing recombinant human GDF-15. In contrast to *in vitro* proliferation and *in vivo* proliferation in immunodeficient NCI nu/nu mice, it is shown that GDF-15 expression mediated a growth advantage in immunocompetent C57BL/6 mice supporting the idea that GDF-15 interferes with the immune system of the tumor host (FIG. 7).

[0143] Further, expression of human GDF-15 rendered anti-PD-1 responsive MC38 colon carcinoma tumors into anti-PD-1 resistant tumors. This was partially reversed by anti-GDF-15 (B1-23)/anti-PD-1 combination treatment, but not by anti-GDF-15 monotherapy and partially by anti-PD-1 alone. (data also shown in FIG. 1)

[0144] Similarly, monotherapy using an anti-GDF-15 antibody showed only minimal improvement, whereas GDF-15 secreting tumors (solid line, FIG. 9B) showed better survival when treated with a combination of anti-GDF-15 B1-23 and anti-PD-1 (FIG. 9). This combination treatment was non-significantly better than anti-PD-1 mono therapy (FIG. 8).

[0145] Since anti-GDF-15 treatment is suggested to increase T cell infiltration, other cancer immunotherapies depending on T cell presence in the tumor should benefit from neutralization of GDF-15.

[0146] Another immunotherapy, anti-CD40 and poly(IC:LC) tumor treatment, which was shown previously to depend on T cell immune responses (van den Boom 2013) was tested in a mouse model with MC38 cells expressing human GDF-15. The combination treatment using GDF-15 neutralization and anti-CD40/poly(IC:LC) resulted in complete rejection of the tumors in 8 out of 10 animals, whereas only 3 out of 10 tumors were cleared by anti-CD40/poly(IC:LC) treatment alone (FIG. 10).

[0147] Although mouse GDF-15 could not be detected in wildtype MC38 cells, which might be due to insufficient analytic assays, a similar experiment comparing isotype control antibody with anti-GDF-15 treatment in combination with anti-CD40/Poly(IC:LC) was done with MC38 cells that were not manipulated to secrete human GDF-15. Similar to the experiment with genetically modified MC38, anti-GDF-15 could improve efficacy of the anti-CD40/Poly(IC:LC) treatment,

however to a lesser extent (FIG. 11).

Secondary Pharmacology (ADCC, CDC)

[0148] GDF-15 is a soluble factor with an interim presence on cells during its maturation. The ability of CTL-002 to bind cell surface associated GDF-15 in vitro implies the possibility that CTL-002 could potentially mediate cell- and complement mediated cytotoxicity to healthy tissue with associated GDF-15.

[0149] ADCC induction was analyzed with ADCC Reporter Assay (Promega). The test and control antibodies were applied in different concentrations to the target cells and were incubated with FcγRIIIa expressing effector cells for 6 h at 37° C. Afterwards a luciferase substrate was added, and the luminescence signal was determined with a luminescence reader after 30 min of incubation at RT.

[0150] As a result, it was found that CTL-002 did not induce any measurable ADCC, while the positive control trastuzumab performed as expected on both cell lines (FIG. 12).

[0151] In addition, the ability of CTL-002 to mediate CDC was also analyzed by measuring C1q binding by ELISA and a cellular reporter assay. As a result, CTL-002 did not show binding to human complement protein C1q, in contrast to the control antibody Rituximab (FIG. 13).

[0152] In a second approach the ability to mediate CDC was tested in a cellular assay. CDC was analyzed on Raji cells with Rituximab as a positive control and on GDF-15 expressing UACC-257 cells. As positive control anti-CD55 and anti-CD59-antibodies were chosen, since neutralization of complement inhibiting molecules was enough to induce CDC, but no antibody directly inducing CDC on UACC-257 was available. As a result, it was found that CTL-002 did not induce any CDC in combination with anti-CD55 and anti-CD59-antibodies, although high levels of the target protein GDF-15 can be detected by flow cytometry (FIG. 14).

[0153] In conclusion, it was surprisingly revealed based on the experimental data that the CTL-002 antibody neither induces ADCC nor CDC.

Pharmacokinetics and Product Metabolism in Animals

[0154] The pharmacokinetics of CTL-002 were analyzed in non-human primates and a PK model of CTL-002 was generated to predict pharmacokinetics in humans. Pharmacodynamic data were generated in the same study (and in the subsequent repeat-dose study in monkeys) by measuring the inhibition of GDF-15 in serum. These datasets were combined into a PK/PD model and used to predict the pharmacodynamic activity of CTL-002 in humans.

[0155] Data from the PK/PD model also provided important information for the estimation of a safe starting dose in the first-in-human study of CTL-002.

Absorption

[0156] A preliminary study has been performed in Cynomolgus monkeys to compare the pharmacokinetics of IgG1- and IgG4-based anti-GDF-15 antibody (CTL-001 and CTL-002, respectively). In this study, each of one male and one female monkey, received a single intravenous injection of 25 mg/kg of either CTL-001 or CTL-002. Blood samples were taken over sixty days.

[0157] The serum concentration-time profiles are shown in FIG. 15 and key pharmacokinetic parameters are summarized in Table 4.

TABLE-US-00005 TABLE 4 Pharmacokinetic parameters of CTL-001 (IgG1) and CTL-002 (IgG4) in monkeys AUC.sub.inf Dose level t.sub.1/2 C.sub.max (μg .Math. Monkey Sex Isotype (mg/kg) (days) (μg/mL) day/mL) 1 Male IgG1 25 9.1 710.9 4514 2 Female IgG1 25 10.6 827.8 5756 3 Male IgG4 25 12.9 949.2 7931 4 Female IgG4 25 14.3 615.2 7980

[0158] In summary, CTL-002 (IgG4 isotype) displayed a slightly longer half-life and slightly higher AUC in the tested monkeys compared to CTL-001 (IgG1 isotype) whilst no adverse effects were observed in any of the animals.

Dose-Response Information and Modelling of Human Therapeutic Doses

[0159] A 2-compartment population pharmacokinetic (PK) model describing the systemic exposure to CTL-002 has been constructed from the non-human primate (NHP) PK data after a single dose

and including trough concentration prior to the third weekly dose.

[0160] There is an accumulation of inactive GDF-15 after administration of CTL-002 and the PK model has been extended to include a binding PD model describing the suppression of GDF-15 (FIG. 16 and FIG. 17). This model describes the observed single dose PK-PD data in NHP and predicts the observed data up to 14 days after repeated weekly administration (2 doses) in the 4 wk GLP toxicity study. There is no evidence to suggest a significant saturable component in the clearance of CTL-002 and maximum target capture can be achieved at doses ≥ 10 mg/kg/wk.

[0161] It should be noted that the observed IgG clearance for CTL-002 in NHP is typical for a human IgG-like molecule in NHP. A possible explanation for the observed over-proportional increase in exposure and loss of GDF-15 capture in some animals after repeated administration of CTL-002 is an ADA response, although this has not been confirmed experimentally (inavailability of ADA-detecting assay in this species).

[0162] The NHP PK-PD model was allometrically scaled to predict the human PK of CTL-002, using exponents of 0.75 for clearance, 0.67 for inter-compartmental exchange and 1.0 for volume. Target binding to NHP and human GDF-15 were also included in the model in order to predict the extent and duration of target suppression (Table 5).

TABLE-US-00006 TABLE 5 CTL-002 - PK-PD model parameters

PK parameter	Unit	NHP
Exponent*	Human	Central Volume L
0.136	1	3.182
Peripheral volume L	0.0866	1
2.021	Exchange coefficient L/day	0.0404
0.67	0.330	CTL-002 clearance L/day
0.0121	0.75	0.128
Elimination rate free GDF-15	1/day	121
-0.25	55.2	Baseline GDF-15 ng/ml
1.25	—	0.5-10
K.sub.D (from PK-PD model)	nM	0.019
**	0.007	*Body weight exponent used to scale from NHP to human (NHP 3 Kg; Human 70 Kg)
**	In vitro	K.sub.D multiplied by 0.18 (observed difference in vitro/in vivo in NHP)

[0163] Given these assumptions, CTL-002 exposure in human was predicted for different dosing regimens (FIG. 18) and the resulting safety margins for C_{max} and AUC were compared with the exposure achieved in the 4 wk GLP toxicity study at 100 mg/kg/Q1 wk (Table 6).

TABLE-US-00007 TABLE 6 Predicted exposure margins: First in human (FIH) study

Dose (mg/kg/Q2wk)	C _{max}	AUC	0.3	683	487	1	186	131	3	54	40	10	14	11	20	6	5
Predicted exposure margins compared with 100 mg/kg CTL-002 in the 4wk GLP toxicity study (NOEL):	First dose C _{max} in human compared with first dose C _{max} in NHP	AUC (steady-state Q2wk) in human compared with AUC (steady-state Q1wk × 2) in NHP	from population PK model; assuming linear PK behavior														

[0164] Observed baseline levels of GDF-15 in the target patient population have been analyzed in a cohort of 34 patients previously treated and being refractory or having relapsed after anti-PD-1 antibody treatment. In this cohort, baseline GDF-15 ranged from 0.35-12 ng/mL. The extent and duration of suppression of systemic GDF-15 is likely to be dependent on the baseline level of GDF-15, with higher production rates of GDF-15 requiring a higher dose to achieve suppression (FIG. 19).

[0165] The estimated FIH dose is a conservative approach, since it will suppress GDF-15 levels below physiologic levels for prolonged periods, but not yet for the full dosing period even in lower level GDF-15 patients. This was done to comply with requests by the consulted agency (PEI, Germany) for the FIH dose. The relationship between baseline GDF-15 level and the extent and duration of target suppression will be explored in the phase I clinical trial.

[0166] The above PK-PD model has been developed to describe the suppression of systemic GDF-15 for various levels of baseline GDF-15. However, the desired target is the tumor micro-environment. If the tumor is largely responsible for the increase in systemic GDF-15 then the amount of GDF-15 in the tumor vasculature should also be considered. Consequently, an estimate of GDF-15 suppression in the tumor micro-environment was included as an extension to the human PK-PD model.

[0167] The serum half-life of GDF-15 in human is predicted to be 18 minutes (allometric scaling

from NHP). Based on this elimination rate, GDF-15 will need to be produced at 1.67 mg/day in order to produce a steady-state serum GDF-15 concentration of 10 ng/mL. For a tumor size of 36 g and an average blood flow rate of 0.2 mL/min/g of tumor tissue (range 0.01-2 mL/min/g—Vaupel 2004) the resulting GDF-15 homodimer concentration in the tumor vasculature will be 161 ng/mL (6.5 nM) i.e. about 16-fold higher than the systemic GDF-15 concentration.

[0168] Based on these assumptions, predictions for tumor GDF-15 suppression in the tumor micro-vasculature are shown in FIG. 20 for various levels of tumor GDF-15. It should be emphasized that predicted tumor GDF-15 is critically dependent on tumor blood flow rate.

[0169] For a patient with a baseline serum GDF-15 level of 2 ng/mL, GDF-15 concentration in the tumor micro-vasculature is predicted to be suppressed to <0.5 ng/mL (average level in healthy individuals) for about 5 days at the proposed starting dose of 0.3 mg/kg. The predicted duration of suppression is much less for patients with a higher serum baseline level of GDF-15 (Table 7).

[0170] Note: suppression of sGDF-15 in the tumor interstitial space, outside the vasculature, has not been incorporated into the PK-PD model since this would require additional assumptions regarding CTL-002 penetration into the tumor environment and GDF-15 levels in the interstitial space.

TABLE-US-00008																																				
TABLE 7 Predicted duration of GDF-15 suppression																																				
GDF-15 baseline	Dose	GDF-15 suppression	Serum GDF-15 suppression	Tumour Serum	Tumour CTL-002	Min	*	Duration	**	Min	* Duration																									
(ng/mL)	(mg/kg)	(ng/mL)	(ng/mL)	(ng/mL)	(days)	(ng/mL)	(days)	0.5	0.5	0.3	0.006																									
>14	0.001	>14	1	0.002	>14	0.000	>14	3	0.000	>14	10	0.000																								
>14	0.000	>14	2	25	0.3	0.024	5.2	4.02	0	1	0.006	>14	0.017	4.2	3	0.002	>14	0.000	>14	10	0.000															
>14	0.000	>14	20	0.000	>14	0.000	>14	10	161	0.3	0.123	0.8	30.964	0	1	0.031	3.8	0.148	0.6	3	0.009	11.8	0.003	8	10	0.002	>14	0.001	>14	20	0.001	>14	0.000	>14	*	Predicted suppression
(lowest concentration) of GDF-15 in either serum or the tumor vasculature																																				
** Predicted duration of GDF-15 suppression below 0.5 ng/mL (mean serum GDF-15 in healthy subjects)																																				

[0171] Moreover, as observed in advanced melanoma patients (10 ng/ml), the proposed starting dose of 0.3 mg/kg CTL-002 will only lower serum GDF-15 slightly below normal levels (0.5 ng/ml) at C.sub.max, at the high end of the range of GDF-15 baseline levels, so that its endogenous functions should not be compromised (FIG. 21).

[0172] In conclusion, CTL-002 is described by a linear PK model. In other words, CTL-002 does not display saturable target-mediated kinetic behavior, as sometimes observed with IgG-like molecules targeted to a membrane receptor.

[0173] In the NHP 4 wk GLP toxicity study, CTL-002 has been shown to be safe and well tolerated at doses which provide adequate exposure margins for clinical testing.

[0174] The proposed FIH starting dose of 0.3 mg/kg/Q2 wk is projected to give a maximum plasma concentration (Cmax) at the end of the 1 hour infusion of 6 µg/mL, which is 683-fold lower than Cmax exposure to CTL-002 at the No-Observed-Effect-Level (NOEL) in NHP. This dose may achieve only transient suppression of GDF-15 in the tumor micro-environment and is considered to be a minimal acceptable biological effect level (MABEL).

[0175] Serum total GDF-15 has been shown to be a useful biomarker of GDF-15 target engagement in NHP and CTL-002 doses ≥10 mg/kg are associated with maintenance of GDF-15 capture (and by inference, GDF-15 suppression). Hence, the total human GDF-15 may be a potential clinical biomarker of GDF-15 target engagement and the FIH clinical study is designed to explore both maximum suppression of GDF-15 for a limited time period and continuous suppression of GDF-15 throughout each dosing cycle.

Toxicology

[0176] CTL-002 is being tested for the treatment of patients with advanced cancer.

[0177] To identify relevant species for non-clinical testing, sequence homology of GDF-15 was compared across species. Sequence homology from humans to Cynomolgus monkeys, mice and rats is 94.6%, 67.9% and 66.1%, respectively.

[0178] CTL-002 binds to a non-linear conformational epitope within GDF-15, illustrated as two boxes in FIG. 22. The sequences of the binding regions of CTL-002 are shown for Cynomolgus monkeys, humans, mice and rats (first four lines from top to bottom).

[0179] The cynomolgus monkey displays 100% sequence homology with the human CTL-002 binding epitope within GDF-15. It is therefore regarded as relevant species for toxicity testing. The binding affinity of CTL-002 to human and cynomolgus GDF-15 is 38.3 pM and 108 pM, respectively.

[0180] A dose response finding (DRF) study of CTL-002 has been performed in rats, as the product was expected to be sufficiently pharmacologically active to achieve sustained complete target inhibition at reasonable intravenous dose levels (binding affinity to rat GDF-15: 449 pM).

However, PK/PD data obtained in the study indicated that CTL-002, even at the highest dose level 100 mg/kg, was not able to saturate GDF-15 binding. Therefore, pivotal toxicology was only assessed in the Cynomolgus monkey in a 4-week study with once weekly intravenous administration of CTL-002. No toxicity was observed up to the highest tested dose of 100 mg/kg.

[0181] A Good Laboratory Practice (GLP)-compliant tissue cross-reactivity study was conducted using human and Cynomolgus monkey tissues. Staining with CTL-002 in the tissue panels examined was limited to the cytoplasm of trophoblasts in the human and monkey placenta, which was consistent with the reported expression of its target protein, GDF-15, in the placenta. No unanticipated cross-reactivity was observed.

Toxicology: Non-Pivotal Studies

[0182] A non-GLP single-dose dose-finding study in Cynomolgus monkeys was performed. The main objective of this study was to support dose selection for the subsequent GLP-compliant 28-day repeat-dose toxicity study. Furthermore, the pharmacokinetics of CTL-002 at various dose levels were characterized.

[0183] One male and one female animal, per dose level (0.1; 1; 10 or 100 mg/kg) were dosed by a single 30-minute intravenous infusion and pharmacokinetic profiles were recorded over 14 weeks. In addition to the pharmacokinetics analysis the occurrence of anti-drug antibodies was assessed pre-dose, as well as 6 and 12 weeks after dosing. GDF-15 and CTL-002 serum levels were quantified, pharmacokinetic data were calculated as total CTL-002 (PK total) and as free CTL-002 (PK free), separately.

[0184] As indicators of toxicity, mortality and clinical signs were checked daily. Bodyweights and food and water consumption were analyzed weekly and body temperatures, ECGs, blood pressure, hematology incl. coagulation as well as clinical chemistry including cytokines were assessed on several occasions during the study.

[0185] None of the animals died prematurely, and there were no CTL-002-related signs of toxicity at any of the tested dose levels.

[0186] The single infusion of 0.1, 1, 10 or 100 mg/kg CTL-002 on test day 1 led to a dose-related increase of the total GDF-15 serum level in all males and females. Target saturation by CTL-002 at dose levels of 10 and 100 mg/kg was indicated by overlapping GDF-15 curves at these two dose levels.

[0187] The measurement of total and free CTL-002 in serum samples obtained up to test day 43 (groups 1 and 2; treatment with 0.1 or 1 mg/kg CTL-002) or test day 99 (groups 3 and 4; treatment with 10 or 100 mg/kg CTL-002) revealed a dose-related exposure of the animals to CTL-002.

[0188] Key pharmacokinetic data are given in Table 8 as means of the one male and one female animal per group.

TABLE-US-00009 TABLE 8 Key pharmacokinetic data obtained in the monkey DRF study CTL-002-TOX-01

Dose level	C.sub.max	AUC.sub.0-t	AUC.sub.inf	Group	[mg/kg b.w.]	[μg/ml]
t.sub.1/2 [h]	[μgh/ml]	[μgh/ml]	Total CTL-002	1	0.1	0.92
				383.4	344.8	412.5
				2	1	21.6
				230.3	4631	
				4855	3	10
				235.8	275.7	77692
				78661	4	100
				2751	352.6	666861
				673354		

[0189] Based on the results of this study, dose levels of 10, 30 and 100 mg/kg were recommended

for the pivotal 4-week repeat dose toxicity study.

Toxicology: Pivotal Studies

[0190] The additional pivotal study was a 4-week repeat-dose toxicity study of CTL-002 in Cynomolgus monkeys (age: 3-4 years) with a 4-week recovery period.

[0191] In this study, CTL-002 was administered by 30-minute intravenous infusions once weekly, i.e. on test days 1, 8, 15, 22 and 29. The recovery period ended on day 58. Dose levels of 0; 10; 30 or 100 mg/kg were administered to 3 male and 3 female monkeys per group plus 2 male and 2 female recovery animals in the control and high dose groups.

[0192] None of the animals died or had to be sacrificed prematurely and no test item-related signs of local intolerance were noted.

[0193] No test item-related effects were noted on behavior and external appearance, the body weight and body weight gain, the food and drinking water consumption, the electrocardiographic parameters and the heart rate, the circulatory functions, the hematological including coagulation parameters, the D-Dimer levels, the clinical chemistry parameters, the cytokine levels, the urinary parameters, the ophthalmological and auditory functions, the organ weights, and the myeloid:erythroid ratio of any of the animals at any dose level. No macroscopic organ changes were noted in any of the animals examined at any tested dose level.

[0194] Histopathology did not reveal any test item-related local or systemic lesions. No test item-related changes were noted during or at the end of the 4-week treatment-free recovery period.

[0195] Based on the above results, the No-Observed-Effect-Level (NOEL) was 100 mg CTL-002/kg by once weekly repeated intravenous 30-minute infusion for 4 weeks, i.e. 5 administrations per animal.

[0196] The C_{max}-levels and AUC-areas for Total CTL-002 revealed a roughly linear dose-related systemic exposure of the animals. No sex-specific differences were noted. An accumulation of Total CTL-002 with time was noted with an accumulation ranging from approx. 2-fold to 6-fold. The calculated mean terminal serum elimination half-lives (t_{sub.1/2}) of Total CTL-002 ranged from 119 to 651 hours.

[0197] Key pharmacokinetic data following the first (days 1-8) and fourth (days 22-29) administration of CTL-002 are given in Table 9 for the male and female animals.

TABLE-US-00010 TABLE 9 Key pharmacokinetic data for Total CTL-002 obtained in the monkey GLP toxicity study CTL-002-TOX-03 after the first and fourth administration

Dose level [mg/kg]	C _{sub.max} [μg/ml]	AUC _{sub.0-t} [μgh/ml]	AUC _{sub.inf} [μgh/ml]	t _{sub.1/2} [h]	Group	b.w.]
2	237	197	21500	475	05	F: 686
10	165	36901	67303	3	30	M: 804
30	292	65026	193062	2292	212	F: 103114
100	2725	312	227335	717624	F: 5428	F: 458
2	1520	486	135788	503693	F: 1030	F: 119
10	4363	308	149037	450025	F: 4935	F: 169
30	12553	651	1388031	9257432	F: 13624	F: 234
100	1329769	F: 3233365				

[0198] Following infusion of CTL-002, GDF-15 serum levels of all animals at all dose levels increased up to 100-1000-fold and remained increased throughout the dosing interval. As indicated in FIG. 23, showing as example the GDF-15 levels in female monkeys throughout the first week after dosing, the increase was comparable across dose groups.

[0199] Therefore, it can be concluded that full target inhibition was obtained at all dose levels and throughout the dosing interval in the pivotal monkey study.

Overall Study Design

[0200] A Phase 1 multi-center, first-in-human (FIH), open-label study consisting of Part A (dose escalation) followed by Part B (expansion) will be performed using the CTL-002 antibody. The main intent of the study is (a) to demonstrate safety of CTL-002 and the combination of CTL-002+anti-PD1/PD-L1 and (b) to demonstrate that patients relapsed post/refractory to anti-PD1/PD-L1 therapy due to elevated GDF-15 will respond again and show tumor shrinkage when the

combination of CTL-002+anti-PD1/PD-L1 is administered.

Part A (Dose Escalation)

[0201] At least 21 subjects will receive in “3+3” cohorts escalating doses of CTL-002 IV given as monotherapy and in combination with an anti-PD-1 checkpoint inhibitor in subjects with advanced-stage solid tumors that relapsed post or were refractory to a prior anti-PD-1/PD-L1 therapy and have exhausted all available approved standard treatments or are not eligible for them anymore. “Backfill cohorts” will recruit additional patients to the highest dose levels.

Part B (Expansion)

[0202] Comprised of up to 5 expansion cohorts of up to 25 subjects per cohort in defined tumor entities that relapsed post or were refractory to a prior anti-PD-1/PD-L1 therapy to further evaluate the safety and efficacy of CTL-002 as monotherapy (one monotherapy cohort to be explored) or in combination with an anti-PD-1 checkpoint inhibitor (up to 4 combination cohorts to be explored) and to confirm the RP2D. The dedicated monotherapy cohort will serve to establish the safety profile of CTL-002 in prolonged monotherapy at a dose considered to be therapeutic (no anti-PD-1/PD-L1 added). Enrollment into all 5 cohorts may occur in parallel.

Treatment Period

Part A (Dose Escalation)

[0203] This study will employ a standard “3+3” dose escalation design for which 3 to 6 subjects will be enrolled at each assigned dose level, per cohort, depending on the occurrence of DLTs.

[0204] The planned doses of CTL-002 to be tested are outlined below: [0205] Cohort 1: 0.3 mg/kg

[0206] Cohort 2: 1.0 mg/kg [0207] Cohort 3: 3.0 mg/kg [0208] Cohort 4: 10 mg/kg [0209] Cohort 5: 20 mg/kg

[0210] The start dose of 0.3 mg/kg for Cohort 1 is fixed. Doses explored in Cohorts 2-5 (as outlined above) may be modified by the Safety Review Committee (SRC) based on emerging data (i.e., available safety, PK/pharmacodynamic, other biomarker data).

[0211] The DLT Observation Period will be the first two treatment cycles (i.e., first 4 weeks) for each dosing cohort. All treatment cycles are defined initially as 2 weeks in duration. CTL-002 will be administered once every two weeks as an IV infusion. Subjects will first receive one dose of CTL-002 given as monotherapy for one cycle, followed by a combination of CTL-002 given together with the defined checkpoint inhibitor for one cycle, where the defined checkpoint inhibitor will be administered at a dose of 240 mg IV given once every 2 weeks.

[0212] For the combination, CTL-002 and the defined checkpoint inhibitor will be given on the same day concomitantly, where CTL-002 will be administered first and for the first combination infusion, there will be a 30-minute observation period to assess safety, which will then be followed by the defined checkpoint inhibitor infusion. The period of observation may be modified (i.e., shortened or lengthened) based on emerging safety data.

[0213] The first two treatment cycles (i.e., first 4 weeks) represent the DLT Observation Period.

[0214] Thereafter, subjects will continue with the combination treatment, until progression or until withdrawal from the study for any other reasons (e.g., toxicity or subject withdraws consent).

[0215] Additional, intermediate dose cohorts may be explored based on emerging data and upon Safety Review Committee (SRC) request. The maximum dose of CTL-002 to be tested in this study will not exceed 20 mg/kg.

[0216] All subjects will be hospitalized overnight after receiving the first dose of CTL-002 and also after receiving the first combination dose of CTL-002 and the defined checkpoint inhibitor, for the purposes of safety observation and to enable logistical collection of sampling time-points (e.g., PK).

Intra-Patient Dose Escalation in Extended Treatment

[0217] If any Cohort 1 subjects are still on 0.3 mg/kg treatment when Cohort 2 has been completed and reviewed by the SRC, the subjects can be increased to the Cohort 2 dose of 1.0 mg/kg.

[0218] If any Cohort 1 or 2 subjects are still on 1.0 mg/kg treatment when Cohort 3 has been

completed and reviewed by the SRC, the subjects can be increased to the Cohort 3 dose of 3.0 mg/kg.

[0219] Note: Any subjects still on 0.3 mg/kg treatment must be treated at 1.0 mg/kg prior to advancing to 3.0 mg/kg in agreement with the Sponsor Medical Monitor.

[0220] The maximum a subject dose can be increased to, through intra-dose escalation, will be 3.0 mg/kg.

[0221] Available safety, PK/pharmacodynamic data, as well as preliminary efficacy data will inform the decision regarding the MTD/dose(s) to be further explored in Part B of the study.

[0222] The MTD is defined as the highest dose level of CTL-002 at which no more than 1 out of 6 subjects experienced a DLT during the first 2 treatment cycles (i.e., the first 4 weeks, where CTL-002 is given as monotherapy [Weeks 1 and 2] and in combination with the defined checkpoint inhibitor [Weeks 3 and 4]).

[0223] In addition, for Cohorts 3-5, in the absence of any DLT, an additional 3 subjects can be recruited into each of these cohorts (up to a total of 6 subjects per cohort), to increase the understanding of the PK and pharmacodynamic data. This occurs while dose escalation continues. These additional “backfill” subjects will receive the combination treatment of CTL-002 and the defined checkpoint inhibitor once every two weeks from Cycle 1 Day 1 onwards, with CTL-002 always administered first and the defined checkpoint inhibitor given thereafter as outlined above.

[0224] For Part A subjects (except backfill subjects), three sequential tumor biopsies are to be taken; one biopsy at baseline, the second biopsy prior to the initiation of the combination therapy (after 2 weeks) and the third after the first cycle of combination therapy (either at the End of Treatment or, if combination treatment is continued, at the end of Cycle 2/beginning of Cycle 3).

[0225] For backfill patients, only two biopsies are mandated; one at baseline, and the second after 4 weeks of combination treatment (either at the End of Treatment or, if combination treatment is continued, at the end of Cycle 2/beginning of Cycle 3).

[0226] These biopsies are mandatory in order to assess immune cell infiltration in the tumor. If a biopsy cannot be taken for safety reasons, this must be discussed with the medical monitor.

[0227] Treatment with CTL-002 in monotherapy (one cycle) as well as in combination with a checkpoint inhibitor (nivolumab) has been safely tolerated up to dose level 5 of CTL-002 (20 mg/kg). No DLT occurred, no grade 4 adverse event in any patient treated. Combination treatment can be started simultaneously, as demonstrated by the so-called backfill cohorts with immediate combination of CTL-002 and anti-PD1/PD-L1 treatment.

[0228] Biomarker analyses show a tumor selective influx of CD8+ and CD4+ cells from DL1-4 consistently. Preliminary analysis indicate that in several patients T cell proliferation is increased in the tumor as demonstrated by CD3/ki-67+ staining. In tumors that at baseline have “cold tumors” characterized by rather low CD8 and CD4 counts are turned into hot tumors by increasing CD8 and CD4 counts.

[0229] First tumor shrinkages have been observed at various dose levels. Most notable is a patient with carcinoma of unknown primary (squamous cell type) treated at dose level 3 backfill cohort that has been relapsed under prior nivolumab treatment and just maintained slowly progressing disease (with stable disease range as per RECIST) when escalated to ipilimumab+nivolumab. Under CTL-002/Nivolumab treatment the patient so far managed to obtain a -49% tumor shrinkage, equal to a confirmed partial remission. Treatment is ongoing. Another patient on dose level 4 with hepatocellular cancer showed -11% tumor shrinkage so far, treatment is ongoing.

Part B (Expansion)

[0230] In Part B of the study, up to 6 cohorts (up to 20 subjects per cohort) each enrolling subjects with a specific tumor type may be enrolled.

[0231] A dedicated CTL-002 monotherapy cohort may be set up in this expansion part of the study to explore the safety profile of CTL-002 given as monotherapy (e.g., in subjects with advanced-stage melanoma). In addition, for this monotherapy cohort, mandatory sequential tumor biopsies

are required to broaden the understanding of the pharmacodynamic effects of CTL-002 in tumor tissue.

[0232] In this monotherapy cohort, two serial tumor biopsies are mandated; one biopsy to be taken at baseline and a second biopsy to be taken after 2 weeks (at the end of Cycle 1/beginning of Cycle 2). All subjects will be treated until progression.

[0233] Then, up to 5 other expansion cohorts of defined tumor populations may be treated with a combination of CTL-002 and the defined checkpoint inhibitor. Tumor indications will consist of PD-1/PD-L1 treatment approved tumor types and subjects that relapsed/progressed on or after anti-PD-1/PD-L1 treatment. Enrollment into expansion cohorts may occur in parallel. All subjects will be treated until progression.

[0234] For the purposes of safety observation and to enable logistical collection of sampling time points (i.e., PK sampling), all subjects will be hospitalized overnight after receiving the first dose of CTL-002 (monotherapy cohort) or after receiving the first combination dose with CTL-002 and the defined checkpoint inhibitor (combination therapy cohort), respectively.

Study Assessments

Tumor Biopsy

[0235] Timing of biopsies are as follows: [0236] Part A (all subjects except backfill subjects): three sequential tumor biopsies are mandated; one biopsy at baseline, the second biopsy prior to the initiation of the combination therapy (after 2 weeks) and the third biopsy after the first cycle of combination treatment (either at the End of Treatment Visit or, if combination treatment is continued, at the end of Cycle 2/beginning of Cycle 3). These biopsies are mandatory in order to assess immune cell infiltration in the tumor. [0237] Part A (backfill subjects): two serial tumor biopsies are mandated; one biopsy at baseline and a second biopsy after 4 weeks (either at the End of Treatment Visit or, if combination treatment is continued, at the end of Cycle 2/beginning of Cycle 3) [0238] Part B (subjects enrolled in the monotherapy cohort): two serial tumor biopsies are mandated; one biopsy at baseline and a second biopsy after the first monotherapy cycle (i.e., 2 weeks).

[0239] There has to be a lesion that is amenable to sequential biopsy, if possible, or a lesion in close proximity, but this lesion should not be the only target lesion that will be radiologically assessed during the course of the study.

[0240] Biomarkers may be analyzed from biopsy tumor tissue samples. Additional immune cell markers and/or tumor markers specific to any of the tumor type may be included.

[0241] Biopsied tumor tissue will be fixed with formalin and embedded in paraffin (FFPE) to determine treatment-induced changes in the number, frequency and spatial location of infiltrating immune cells including but not limited to leukocytes, different lymphocytes (e.g., CD4+ and CD8+ T cells, B cells, NK cells) by histology before and after treatment with CTL-002 or in combination with the defined checkpoint inhibitor. Moreover, the expression of the CTL-002 drug target, GDF-15 protein and mRNA, will be determined.

Tumor Lesions

[0242] For subjects in Part A of the study, the target cutaneous lesions selected for RECIST evaluation will be measured by caliper and photographed. In addition, the number of cutaneous lesions will be recorded. For clinical measurements of cutaneous lesions in Part A, documentation by color photography (including size measurement) and caliper measurement of lesion will be performed at Baseline within -7 days before infusion of CTL-002, every 4 weeks during extended treatment, at End of Treatment Visit and during follow-up.

Assessment of Safety

[0243] The safety and tolerability of IV infusions of CTL-002 monotherapy and CTL-002 in combination with the defined checkpoint inhibitor will be evaluated by the incidence of AEs (all AEs will be evaluated according to NCI CTCAE v5.0), SAEs, DLTs, and use of concomitant medications. Safety assessments will include: ECGs, physical examinations including neurological

examination to exclude motor neuropathy, ECOG performance status, vital signs and clinical laboratory samples (hematology, clinical chemistry, coagulation, thyroid function (thyroid stimulating hormone [TSH] and free T3), cytokines, assessment of hemoglobin A1c [HbA1c], N-terminal B-type natriuretic peptide [NT proBNP], and urinalysis).

[0244] Subjects are assessed for safety at Screening, as well as during treatment until the Safety Follow-up Visit. Thereafter safety related to the study is further captured during the follow-up of 12 months (Part A)/24 months (Part B) post-treatment.

Vital Signs

[0245] Vital signs including systolic and diastolic BP (sitting), pulse rate, temperature, respiratory rate, and oxygen saturation should be evaluated. Additional vital sign measurements may be performed if clinically warranted.

Physical and Neurologic Examination

[0246] A physical examination will be performed at Screening and will include examination of head, eyes, ears, nose, throat, neck, cardiovascular, chest/lungs, abdomen (including liver and spleen size), extremities, skin, and lymph nodes, as well as a brief neurologic examination to assess motor neuropathy.

[0247] Additional physical examination assessment time points are outlined below and in the SoAs in Table 5, Table 6 and Table 7.

Performance Status

[0248] Performance status will be assessed at Screening according to ECOG criteria as follows:

[0249] 0=Fully active, able to carry out all pre-disease activities without restrictions [0250]

1=Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature (e.g., light housework, office work) [0251] 2=Ambulatory and capable of self-care, but unable to carry out any work activities. Up and about more than 50% of waking hours.

[0252] 3=Capable of only limited self-care, confined to bed or chair more than 50% of waking hours [0253] 4=Completely disabled, cannot carry on self-care, totally confined to bed or chair

[0254] 5=Death

Cardiac Function Monitoring

Cardiac Function Monitoring

[0255] Subjects will undergo a thorough monitoring for cardiac/vascular AEs and protective measures are in place to exclude subjects at risk from trial participation.

[0256] At baseline, subjects undergo an ECG, an echocardiography (or MUGA if ECHO cannot be performed) and testing for N-terminal pro b-type Natriuretic Peptid levels (NT-proBNP, heart-failure screening). Testing for NT-proBNP will be repeated every 2 weeks for 3 months and thereafter monthly or in case of any suspicion regarding cardiac/vascular damage of any type (then combined with ECG and echocardiography, again).

[0257] A single, 12-lead ECG will be performed.

[0258] All ECG monitoring is to be performed locally at the Investigator site.

[0259] The subject should be relaxed and in a recumbent or semi-recumbent position at least 5 minutes before recording an ECG.

[0260] Additional ECG testing may be performed at the Investigator's discretion if deemed clinically warranted.

Clinical Laboratory Assessment

[0261] Samples for laboratory testing listed as below will be collected. All tests are performed locally.

[0262] Hematology/coagulation, clinical chemistry results must be available and reviewed and deemed acceptable by the Investigator or authorized designee, prior to CTL-002/PD-1/PD-L1 administration.

[0263] Clinically significant abnormal tests must be repeated to confirm the nature and degree of the abnormality. When necessary, appropriate ancillary investigations should be initiated. If the abnormality fails to resolve or cannot be explained by events or conditions unrelated to the study

medication or its administration, the Medical Monitor must be consulted.

[0264] The clinical significance of an abnormal test value, within the context of the disease under study, must be determined by the Investigator which includes significant shifts from baseline within the range of normal that the Investigator considers to be clinically important.

TABLE-US-00011 TABLE 10 Safety Laboratory Testing Hematology Hemoglobin, hematocrit, RBC, WBC with differential, ANC, AMC, platelet count Clinical CRP, creatinine, LDH, calcium, electrolytes (sodium and Chemistry potassium), total bilirubin, GGT, albumin, alkaline phosphatase, AST, ALT, and glucose Coagulation Includes: α PTT, PT/INR INR/PT should be measured daily for any subject experiencing ALT or AST elevations $\geq 3 \times$ ULN with concomitant elevation in bilirubin $\geq 2 \times$ ULN until resolution to baseline of the liver function test abnormality.

AT III D-Dimer Serology Analysis for HIV1 and HIV2, HBV, HCV, TBC, SARS- COV-2

Urinalysis pH, ketones, specific gravity, bilirubin, protein, blood and glucose will be assessed by dipstick. α PTT = activated partial prothrombin time; AMC = absolute monocyte count; ALT = alanine aminotransferase; ANC = absolute neutrophil count; AST = aspartate aminotransferase; AT III = Antithrombin III; CRP = C-reactive protein; GGT = gamma-glutamyltransferase; HBV = hepatitis B virus; HCV = hepatitis C virus; HIV1 = human immunodeficiency virus 1; HIV2 = human immunodeficiency virus 2; INR = international normalized ratio; LDH = lactate dehydrogenase; PT = prothrombin time; RBC = red blood cell count; TB = tuberculosis; ULN = upper limit of normal; WBC = white blood cell count

Pharmacokinetics

[0265] The PK of CTL-002 given as monotherapy and/or in combination with the defined checkpoint inhibitor will be measured from blood samples collected at the start of treatment and at various subsequent time points (Part A). Additional PK data may be evaluated in the expansion groups (Part B).

[0266] Blood samples will be taken at the start of treatment and at various subsequent time points to determine, if antibodies directed against CTL-002 may have developed.

Monitoring of Systemic Cytokines/Chemokines (Pharmacodynamics)

[0267] Serum samples will be collected for measurement of cytokines, chemokines and other circulating biomarkers to assess pharmacodynamic effects as well as safety. Cytokines and Chemokines to be analyzed may include but are not limited to: tumor necrosis factor alpha (TNF- α), interferon (IFN)-custom-character, interleukin (IL)-1 β , IL-2, IL-4, IL-6, IL-8, IL-10, IL-12p70, IL-13, CXCL9 (monokine induced by gamma [MIG]) and CXCL10 (IP-10).

Exploratory Assessments

[0268] Serum biomarker testing on specimens specifically collected for future biomedical research during this clinical trial (retention aliquots) might be conducted to identify serum factors (e.g., but not limited to metabolites, soluble growth factors, cytokines, chemokines) important for anti-GDF-15 (CTL-002) therapy. Retrospective biomarker studies will be conducted with appropriate biostatistical design and analysis and compared to PK/Pharmacodynamic results, previously assessed biomarkers or clinical outcomes.

Efficacy Assessment: Imaging Assessments (Local Testing)

[0269] Tumor response is evaluated according to institutional standards using RECIST V1.1 as well as imRECIST criteria. For the purposes of this study, subjects will undergo evaluation at Screening for a baseline scan and should be re-evaluated every 8 weeks beginning at Cycle 3 and/or from the End of Treatment Visit, then after this time response assessments may be performed as per local institutional guidelines until the end of the Efficacy and Survival Follow-up.

[0270] All lesions identified at Screening/baseline will be consistently followed using the unique lesion number assigned at Screening/baseline and need to be documented in the subject file and in the eCRF.

[0271] The same method of assessment (imaging modality, e.g., MRI, CT) must be used to characterize each identified and reported lesion at baseline and during all follow-up examinations

for an individual subject. If there is a change in modality, then the trial site may be asked to explain the reason for the change in the eCRF. A change in modality may be considered a protocol deviation.

[0272] Each efficacy time point/visit may be completed up to a window of ± 7 days.

[0273] A central reading of the images by a reading center will be performed post-hoc in addition to local reading by the investigator during the trial.

Definition of Progressive Disease According to RECIST V1.1 and imRECIST:

[0274] RECIST assessments will be used to identify subjects with possible progression of disease (Eisenhaur et al, 2009). As defined by modified RECIST V1.1 criteria for immune-based therapeutics or imRECIST criteria (Hodi et al, 2018), the date of initial potential progression by RECIST scanning will be defined as the immune unconfirmed progressive disease (iUPD) date. Subjects with an iUPD date who are stable will continue to participate in the study as planned and be reassessed for progression 4 to 8 weeks after the initial assessment. If the confirmatory assessment supports PD, the date of disease progression will be the iUPD date. If the confirmatory assessment does not support PD, the subject does not have disease progression and the iUPD date is ignored; such subjects will remain in the study as planned and continue the next imaging evaluation as planned per protocol.

[0275] Anti-tumor activity will be assessed per Investigator assessment using RECIST V1.1 and the Immune Response Criteria according to imRECIST as described below. [0276] Contrast CT scans of the chest, abdomen and pelvis, MRI or positron emission tomography-computed tomography (PET-CT). [0277] Disease response and disease progression will be evaluated in this study using RECIST and imRECIST criteria. [0278] Subjects with brain and/or leptomeningeal metastases that are symptomatic or untreated or that require current therapy will not be eligible for the study. Brain imaging must not be older than 12 weeks. Results with abnormal/unexpected findings of brain MRI should be discussed with the Medical Monitor as part of the screening process. [0279] The same method of assessment and the same technique should be used to characterize each identified and reported lesion at baseline and during follow-up. If there is a change in modality, then the trial site may be asked to explain the reason for the change in the eCRF. A change in modality may be considered a protocol deviation.

Results

[0280] The study was initiated in December 2020 and enrolled the first patient on Dec. 9, 2020. Cohorts 1-4 have been completed without dose-limiting toxicity (DLT) and dose escalation continues. Note: Interim data.

Patients and CTL-002 Treatment:

[0281] This interim report includes demographics and preliminary safety data of the first 16 patients treated in the CTL-002-001 trial.

TABLE-US-00012 TABLE 11 Patient demographics per dose cohort

	Dose Level 1	Dose Level 2	Dose Level 3	Dose Level 4	Dose Level 5
(0.3 mg/kg)	(1.0 mg/kg)	(3 mg/kg)	(10.0 mg/kg)	(20.0 mg/kg)	Total N = 3
N = 4*	N = 6	N = 6	N = 3	N = 22	Age, years, 72 (66-77)
56 (54-61)	59 (54-65)	66 (34-79)	62 (61-65)	61.5 (34-79)	median (range)
Gender male, number (%)	2 (67)	1 (25)	1 (17)	6 (100)	2 (67)
12 (55)	female, number (%)	1 (33)	3 (75)	5 (83)	0 (0)
1 (33)	10 (45)	ECOG Baseline	0 (0)	3 (75)	4 (67)
3 (50)	2 (67)	8 (50)	1 (33)	8 (50)	Tumor type
Pleural	Oropharyngeal	CA	Melanoma	Renal	CA
Melanoma	Melanoma	Melanoma	TN-Breast	CA	Ovarian
CA	Hepatocellular	CA	Uveal	Colorectal	CA
Melanoma	Squamous	CA	NSCLC	Melanoma	NSCLC
Melanoma	Tbd.	Ocular	Cholangiocellular	Tbd.	melanoma
CA	Tbd.	Prior lines of therapy	Number (average %)	11 (3.7)	15 (3.8)
28 (4.7)	23 (3.8)	13 (4.3)	91 (4.1)	1-3	2 (67)
2 (50)	3 (50)	4 (67)	1 (33.3)	12 (54.5)	4-5
0 (0)	2 (50)	1 (17)	1 (16.5)	1 (33.3)	5 (22.7)
≥6	1 (33)	0 (0)	2 (33)	1 (16.5)	1 (33.3)
5 (22.7)	*1 patient replaced due to delayed start of combination treatment				

Preliminary Safety and Tolerability of CTL-002:

[0282] So far monotherapy as well as the combination with nivolumab showed excellent tolerability. No dose-limiting toxicity (DLT) has occurred and no safety event of concern has been observed to date. Overall, a total of 111 adverse events have been reported for dose levels 1-5 so far. Three of the AEs were classified as SAE and as at least possibly related to CTL-002 (two due to prolonged hospitalization, one declared to be as important medical event per Investigator assessment; all Common Terminology Criteria for Adverse Events (CTCAE) grade 1-2).

[0283] No CTCAE Grade ≥ 4 was observed.

[0284] In summary, the side effect profile is very mild.

Biomarker Strategy and Analyses

[0285] This clinical trial explores serum- and tissue-based biomarkers. Apart from classic immune-system activation markers such as serum cytokines, specific analyses are conducted to evaluate the immunomodulatory effect of GDF-15 in the tumor microenvironment. Among other parameters, baseline GDF-15 levels, intratumoral GDF-15 levels as well as numbers and profiles of tumor-infiltrating leukocytes are analyzed prior to and under GDF-15 neutralization by CTL-002.

Substantial tumor-selective GDF-15 expression was confirmed for most tumors analyzed. Tumor and tumor-stroma selective influx of mainly CD8⁺ and CD4⁺ T cells under CTL-002 dosing was observed in the majority of patients, with effect being seen from dose level 1 onwards. Preliminary analysis indicate that in several patients T cell proliferation is increased in the tumor as demonstrated by CD3/ki-67⁺ staining. In tumors that at baseline have “cold tumors” characterized by rather low CD8 and CD4 counts are turned into hot tumors by increasing CD8 and CD4 counts.

[0286] Furthermore, some tumors showed reactive PD-L1 upregulation, which is an indirect sign of IFN-gamma release.

TABLE-US-00013 TABLE 12 List of analytes in blood, tumor and urine for PD/PK modelling, immunological assessment and biomarker identification. Methods/Materials Blood Serum Tumor Urine GDF-15 protein (ELISA, ECL) – + – + PK total CTL-002 (ECL) – + – – ADA (ECL) – + – – Cytokines/Chemokines (ELISA, ECL) – + – – IFN-g/IL-1b/IL-2/IL-4/IL-6/IL-8/IL-10/IL-12p70/IL13 CXCL9/CXCL10 Tumor infiltrating immune cells/contexture – – + – (IHC) CD3/CD4/CD8/Foxp3/Ki67/GranzB/ CD11c/CD29/CD68/MHCII/CKSox10 PD-L1 GDF-15 (IHC to proform) Other markers optional Tumor GDF-15 mRNA (FISH) (optional) – – + – Molecular biomarkers (+) – + – Nanostring nCounter PanCancer IO 360™ Panel Mutational load (optional) Other molecular biomarkers optional (MSI, mutations) Sample biobanking for optional analyses + + + (on request)

Preliminary Response Assessment

[0287] First tumor shrinkages have been observed at various dose levels. Most notable is a patient with carcinoma of unknown primary (squamous cell type) treated at dose level 3 backfill cohort that has been relapsed under prior nivolumab treatment and just maintained slowly progressing disease (with stable disease range as per RECIST) when escalated to ipilimumab+nivolumab. Under CTL-002/Nivolumab treatment the patient so far managed to obtain a –49% tumor shrinkage, equal to a confirmed partial remission. Treatment is ongoing. Another patient on dose level 4 with hepatocellular cancer showed –11% tumor shrinkage so far, treatment is ongoing.

CONCLUSIONS

[0288] Recent preclinical data by us and others indicate that GDF-15 potently (1) prevents T cell infiltration into the tumor microenvironment (TME) and that it (2) suppresses a potent immune response within the (TME) by other mechanisms, too. GDF-15 thus plays a key role in suppressing effective anti-tumor immune responses.

[0289] The GDFATHER (GDF-15 Antibody-mediated Effector cell Relocation) phase 1 trial explores the safety, PK and PD and preliminary antitumoral activity of the GDF-15 neutralizing antibody CTL-002 in monotherapy and combination with a checkpoint-inhibitor (CPI) in CPI-relapsed/-refractory patient populations. Anti-cachexia effects are investigated, too.

[0290] Dose level 1-5 have been completed safely with excellent tolerability and no DLT.

[0291] The preliminary pharmacodynamic analyses from sequential tumor biopsies (dose level 1-4) indicate a CTL-002-mediated selective T cell shift into the tumor microenvironment.

[0292] Preferred doses and dosage regimens for the antibodies of the invention including CTL-002 are 3, 10 or 20 mg/kg/Q2 wk; a more preferred dose and dosage regimen is 10 mg/kg/Q2 wk. As reflected by the above-described favorable clinical effects which were observed in patients at dose levels 3 and 4 so far, it is expected that these dose levels are highly effective. Additionally, the dose and dosage regimen selection is based upon extensive investigations carried out by the inventors, including pharmacomodelling and obtained PK/Pharmacodynamic data from Part A indicating complete GDF-15 neutralization at this dose in patients with all ranges of baseline GDF-15 serum levels. A review of all treatment doses and their adverse events, the PK/Pharmacodynamics observed so far, and a thorough pharmacomodelling exercise conducted indicates full GDF-15 suppression within the tumor microenvironment at 10 mg/kg of CTL-002 even at elevated baseline serum concentrations of GDF-15 of up to 10 ng/ml GDF-15 in serum, corresponding to approximately 160 ng/ml GDF-15 in immediate tumor proximity. No safety events of concern have been observed so far at this dose and no DLT occurred at this dose and there is still a >10-fold safety margin compared with the NOAEL in Non-human primates (NHP). Thus, it is expected that the above-indicated preferred doses are particularly effective and safe.

[0293] The above-indicated observations with regard to efficacy, safety, and PK/Pharmacodynamic data also show that it will be possible to advantageously administer the anti-GDF-15 antibody at a preferred dose of between 10 and 20 mg/kg, more preferably 20 mg/kg, and at a dosage regimen of at least one administration cycle, wherein the cycle is a period of four weeks and wherein said dose is to be administered at least once (i.e. preferably once) in each of the at least one cycle. This dosage regimen has a longer administration cycle of four weeks, but the preferred dose of between 10 and 20 mg/kg, more preferably 20 mg/kg, will allow to obtain an advantageous safety and efficacy profile similar to the preferred 10 mg/kg/Q2 wk regimen and is compatible with the observed PK/Pharmacodynamic profile.

[0294] Similarly, the above-indicated observations with regard to efficacy, safety, and PK/Pharmacodynamic data also show that it will be possible to advantageously administer the anti-GDF-15 antibody at a preferred dose of between 10 and 20 mg/kg and at a dosage regimen of at least one administration cycle, wherein the cycle is a period of three weeks and wherein said dose is to be administered at least once (i.e. preferably once) in each of the at least one cycle. This dosage regimen has a longer administration cycle of three weeks, but the preferred dose of between 10 and 20 mg/kg will allow to obtain an advantageous safety and efficacy profile similar to the preferred 10 mg/kg/Q2 wk regimen and is compatible with the observed PK/Pharmacodynamic profile.

[0295] The above findings indicate that a treatment with anti-GDF-15 antibodies according to the invention can provide a considerable clinical benefit for cancer patients. Importantly, this benefit is also observed in patients who had previously been refractory to some of the most advanced treatment options such as therapy with antagonists of the PD-1/PD-L1 axis (e.g., nivolumab).

In Silico Determination of Potential Stability Risks in Antibody with IgG1 Backbone “H1L5”

[0296] In a first step to approach the goal to provide a stable formulation for the inventive antibody, Applicants set out to determine which parts and sequences of the antibody were potentially at risk in the future formulation effort. To do so, an in silico determination was done. The humanised anti-GDF-15 antibody H1L5 was screened with in silico manufacturability assessment tools. The amino-acid sequence of H1L5, composed of a full-length Kappa isotype light chain and a full length IgG1 heavy chain, was screened for the sequence motifs and features of a number of potential developability issues and for aggregation risk. It was shown that H1L5 has a potential CDR deamidation site and an oxidation site that could benefit from in vitro evaluation. The antibody also has other issues in the form of potential oxidation and acid-labile sites as well as C-terminal clipping.

[0297] Therapeutic proteins are complex and very heterogeneous due to post-translational modifications (PTMs) and chemical modifications. These modifications include glycosylation, deamidation, oxidation and variations of N- and C-termini. Modifications which result in relevant product-related variants are classified as critical quality attributes (CQAs) by regulators. CQAs are given narrow acceptance criteria and their variations are monitored by appropriate qualitative and quantitative methods.

[0298] Modifications can be attributed to the host cell system, manufacturing processes and storage conditions. They can either relate to the chemical stability of the molecule or the intrinsic physical stability in the form of aggregation potential. Aggregation is an issue which has such a potential impact on safety, quality and efficacy that one or more CQAs are generally defined for it.

[0299] Protein aggregation is a commonly encountered problem during biopharmaceutical development. It has the potential to occur at several different steps of the manufacturing such as fermentation, purification, formulation and storage. The potential impact of aggregation spans not only the manufacturing process but also the target product profile, delivery and, critically, patient safety.

[0300] Aggregation depends on the protein itself (intrinsic aggregation propensity) and on environmental factors such as pH, concentration, buffers, excipients and shear-forces. However, the fundamental difference as to why one antibody aggregates during a process step or during manufacturing and others do not is encoded in the antibodies' amino-acid sequences and their intrinsic aggregation propensities. Aggregation poses a risk to safety, quality and efficacy of antibodies.

[0301] Asparagine deamidation is a non-enzymatic reaction that over time produces a heterogeneous mixture of asparagine, iso-aspartic acid and aspartic acid at the affected position. Deamidation is caused by hydrolysis of the amide group on the side-chains of asparagine and glutamine. Whilst glutamine deamidation may occur in therapeutic proteins the manufacturability focus is on asparagine deamidation. Three primary factors influence the deamidation rates of peptides: pH, high temperature and primary sequence. The secondary and tertiary structures of a protein can significantly alter the deamidation rate. In addition to causing charge heterogeneity, asparagine deamidation can affect protein function if it occurs in a binding interface such as in antibody CDRs. Deamidation has also been reported to cause aggregation.

[0302] Aspartic acid isomerisation is the non-enzymatic interconversion of aspartic acid and iso-aspartic acid residues. The peptide bond C-terminal to aspartic acid can be susceptible to fragmentation in acidic conditions. As these reactions proceed through intermediates similar to those of the asparagine deamidation reaction; the rate of aspartic acid isomerisation and fragmentation is influenced by pH, temperature and primary sequence. Aspartic acid isomerisation can affect protein function when it occurs in binding interfaces such as antibody CDRs. Isomerisation also causes charge heterogeneity and can result in fragmentation caused by cleavage of the peptide back-bone. The fragmentation reaction primarily occurs below pH 5 and Asp-Pro peptide bonds are more labile than other peptide bonds. Aspartic acid isomerisation has the potential to increase immunogenicity, a risk that is further increased as fragmentation favours the occurrence of aggregates.

[0303] C-terminal Lysine processing is a modification in antibodies and other proteins that occurs during bioprocessing likely due to the action of basic carboxypeptidases. C-terminal lysine processing is a major source of charge and mass heterogeneity in antibody products as species with two, one or no lysines can be formed.

[0304] The isoelectric point (pI) of a protein is the pH at which the protein has zero net electrical charge. The isoelectric point is dependent on the number and type of charged residues in the protein, their spatial arrangement and degree of solvent accessibility. The prediction of the isoelectric point from the amino-acid sequence assumes a denatured protein. While it is known that predicted and measured isoelectric points differ, a relationship between the two values can be seen.

When a protein solution is at a pH equal to the pI of the protein the repulsive electrostatic forces between charges on the protein molecules are minimised. The lack of repulsive electrostatic forces may increase the risk of hydrophobic surface patches becoming aggregation hot-spots.

[0305] N- and O-Glycosylation is a post-translational modification appearing in therapeutic proteins such as antibodies, blood factors, EPO, hormones and interferons]. The attachment of the carbohydrate to amino acid residues occurs at the side chain nitrogen atom of Asparagine in N-Glycosylation and the side chain oxygen atom of Serine and Threonine in O-linked glycosylation. Some immunoglobulin V-genes contain Asparagine residues in the CDRs which may result in an N-glycosylation motif forming during selection, with approximately 20% of all antibodies being glycosylated in the variable regions in vivo. Proper glycosylation is important not only for folding, but also stability, solubility, potency, pharmacokinetics and immunogenicity. Unintended glycan structures in or near binding interfaces such as CDRs may occlude the binding region or introduce steric hindrance thereby reducing binding affinity. Glycan structures can vary in branching and composition thereby introducing further heterogeneity which may have to be characterised and controlled.

[0306] Oxidation: Several amino acids are susceptible to damage by oxidation caused by reactive oxygen species (ROS), amongst them are histidine, methionine, cysteine, tyrosine and tryptophan. Oxidation is generally divided into two categories: site-specific metal catalysed oxidation and non site-specific oxidation. Methionine and to a lesser extent tryptophan are more susceptible to non site-specific oxidation. While methionine is primarily sensitive to free ROS, tryptophan is more sensitive to light induced oxidation. The degree of sensitivity is determined in part by the solvent accessibility of the side chain; buried residues are less sensitive or take longer to react.

[0307] Pyroglutamate formation is a modification occurring in proteins with an N-terminal glutamine or glutamic acid residue, where the side chain cyclises with the N-terminal amine group to form a five-membered ring structure. As many antibody light and heavy chains have an N-terminal Glutamine or Glutamic acid residue, pyroglutamate formation is a common modification, especially for sequences with an N-terminal glutamine. N-terminal cyclisation causes mass and charge heterogeneity which has to be controlled and monitored. Pyroglutamate formation is commonly found in antibodies with an N-terminal Glutamine. Glutamic acid to pyroglutamate conversion is unlikely to pose a safety risk, however the N-termini in antibodies are proximal to CDRs and the charge variation may influence binding affinity.

Abbreviations

[0308] CDR Complementarity Determining Region [0309] Fc Fragment crystallisable of an antibody [0310] FR Framework Region [0311] FRx Framework Region x (FR1, FR2, FR3, FR4) [0312] Hx CDR x, Heavy chain (H1, H2, H3) [0313] H:Ala11 Heavy chain Alanine at ordinal position 11 [0314] IgG Immunoglobulin G antibody [0315] Lx CDR x, Light chain (L1, L2, L3) [0316] L:Ala11 Light chain Alanine at ordinal position 11 [0317] pI Isoelectric point [0318] VH Variable domain, Heavy chain [0319] VL Variable domain, Light chain

Results

Sequence Liability Maps

[0320] The two sequence maps shown in FIG. 24 and FIG. 25 show diagrammatic representations of the locations of all identified sequence liabilities. In these diagrams the locations of the domain boundaries and the CDRs are indicated in relation to the overall sequence. A scheme is used to indicate the type of liability detected at a given position. Asparagine residues predicted to potentially be involved in both deamidation and N-glycosylation are indicated using both schemes. All indicated residues in both FIGS. 24 and 25 represent potential issues for development of a stable formulation for the present antibody.

[0321] This section focusses on the predicted aggregation and the most important developability issues for H15.

[0322] The result of the antibody aggregation risk prediction is given in Table 13.

[0323] The potential developability issues are summarised in Table 14 summarizing sites which might be particularly problematic for formulation design during development.

TABLE-US-00014 TABLE 13 Aggregation Prediction Results Comment Low risk The antibody is predicted to have some aggregation risk

TABLE-US-00015 TABLE 14 Potential Issues for Development Amino- acid Chain and Position Comment L (light) L: Asn92 CDR L3 Asparagine with deamidation potential H (heavy) H: Met34 CDR H1 Methionine with oxidation potential H H: Trp55 Buried CDR H2 Tryptophan with oxidation potential H H: Asp74 Acid labile Aspartyl-Proline peptide bond, potential backbone cleavage site. The modification site is located in the FR3 (framework 3 region) and is unusual H H: Asp88 Acid labile Aspartyl-Proline peptide bond, potential backbone cleavage site. The modification site is located in FR3 and is unusual H H: Met105 CDR H3 Methionine with oxidation potential H H: Lys448 Chain susceptible to heterogeneity from C-terminal cleavage [0324] Clearly, there were several risk factors which could potentially destabilize the antibody during further formulation efforts.

[0325] In a second step, the antibody H1L5 was engineered to an IgG4 backbone, as described herein elsewhere, and was then designated as CTL-002. With the IgG4 backbone three of the above identified risk factors could be eliminated, namely [0326] 1) the K448 of IgG1 has been deleted [0327] 2) the N at position 204 of IgG1 has been replaced by a D in the IgG4 antibody [0328] 3) the S at position 132 of IgG1 has been replaced by a C in the IgG4 antibody

[0329] The change from IgG1 to IgG4 has thus eliminated three potential risk factors for the provision of a stable antibody formulation.

Thermal and Colloidal Stability Testing

[0330] Based upon the knowledge obtained with the in silico experiment carried out above, the inventors decided on first basic formulations which might be suitable to stabilize the specific antibody structure with issues as found in the in silico experiment carried out above.

[0331] While it was considered clearly preferable to have a stable liquid formulation it was decided that it would be necessary to also include a potential lyophilized formulation in view of potential stability problems with this antibody in liquid surroundings. On that basis, it was decided that it would be necessary to test [0332] Four liquid formulations [0333] One lyophilized formulation [0334] Two different pH points, both of which were chosen to be far below the isoelectric point of the antibody [0335] Two different buffer systems [0336] Several different excipients in different concentrations

[0337] The pH points were chosen after giving specific thoughts to the determination of the pI of the antibody and taking into account potential best surroundings to reduce aggregation and to increase repulsive electrostatic forces between protein molecules for the present particular antibody.

[0338] This led to the following study design:

[0339] pH and ionic strength screening were performed at microliter scale with selective analytics. The thermal and colloidal stability was studied by dynamic light scattering analysis and intrinsic fluorescence, light scattering analysis and micro calorimetry respectively.

TABLE-US-00016 TABLE 15-1 Formulation conditions of the pH and ionic strength screen Formulation ID Buffer pH Excipients F1 20 mM histidine 6.0 240 mM sucrose F2 20 mM histidine 5.5 240 mM sucrose F3 20 mM histidine 6.0 150 mM sucrose, 50 mM arginine-HCl F4 20 mM histidine 6.0 150 mM NaCl F5 20 mM Na-citrate 6.0 240 mM sucrose

TABLE-US-00017 TABLE 15-2 Analytical testing Sample Parameter Analytical method concentration Melting temperature Differential scanning 5 mg/ml micro calorimetry Unfolding temperature Intrinsic fluorescence 10 mg/ml Onset of aggregation Static light scattering 10 mg/ml Dissociation constant Dynamic light scattering 1-10 mg/ml

Materials

[0340] Drug substance (DS) of CTL-002 was processed and provided by Lonza Biologics Plc,

Slough (UK). The DS batch was shipped and stored at 2-8° C. from the date of arrival to the date the material was aliquoted and used for different studies.

TABLE-US-00018 TABLE 16-1 Drug substance information as delivered by manufacturing site
Drug substance Details Name CTL-002 Batch Number L35404/H27 (pool from CT08 clone)
Concentration 24.9 mg/ml Buffer 20 mM histidine, pH 6.0 Storage conditions 2-8° C. Extinction coefficient 1.428 ml .Math. mg.sup.-1 .Math. cm.sup.-1 at 280 nm Molecular weight 144 539 Da
TABLE-US-00019 TABLE 16-2 List of chemicals Material No Material name (supplier) Supplier name (supplier) L-Histidine J. T. Baker 2080-06 L-Histidine, Monohydrochloride, J. T. Baker 2081-06 Sucrose Pfanstiehl S-124-2-MC Arginine-HCl Sigma A5131-500G Sodium Chloride Sigma S7653-1KG Trisodium citrate dihydrate Merck 1.37042.1000 Citric acid monohydrate Sigma C1909-500g Sodium hydroxide solution 50% Honeywell Fluka 71686-1L Purified water In-house n/a

TABLE-US-00020 TABLE 16-3 Main consumables Material name Supplier name Material Nr (supplier) Vivapsin 6 10,000 MWCO PES Sartorius VS0602

TABLE-US-00021 TABLE 16-4 Main equipment Equipment name Model Manufacturer Analytical balance XPE206 BR Mettler Toledo Centrifuge 5810R Eppendorf pH meter 780 Metrohm UV/VIS SoloVPE C Technologies Inc Photospectrometer Capillary Differential MicroCal VP Malvern Scanning Calorimetry Dynamic Light DynaPro Plate Wyatt Technology Scattering Reader II Fluorescence & Static UNit Unchained Labs light scattering Reader

Methods

Preparation of Test Material

[0341] DS was subjected to buffer exchange in formulations F2 and F3 by using centrifugal concentrators. The protein concentration was monitored at the end of processing.

[0342] In all formulations, the protein concentration was adjusted by addition of the specific formulation buffer.

Results

Thermal Stability

[0343] The thermal structural stability of a protein can be assessed by the temperature at which protein aggregates (aggregation onset temperature (Tagg)) as well as by the temperature at which it unfolds from the native (folded) state to a denatured (unfolded) state. The mid-point of the unfolding transition, which is defined as the temperature at which there is an equal population of folded and unfolded proteins in solution, is termed melting temperature (Tm) when assessed by traditional DSC measurements, and unfolding temperature (Tunfold) when assessed by intrinsic fluorescence. The unfolding of IgG molecules presents two or three transitions reflecting the unfolding of Fab and Fc (CH2 and CH3) fragments.

[0344] Onset of aggregation was determined by light scattering at approx. 61-68° C. for the five samples. The Tagg values can be ranked as follows: F5>F1-F4>F3>F2

[0345] Unfolding temperatures were observed at temperatures of approx. 64-68° C. and melting temperatures at temperatures of approx. 65-69° C. Both methods gave a similar formulation ranking: F5>F1-F3>F4>F2.

[0346] Overall, the thermal stability is higher at pH 6.0 than at pH 5.5. NaCl lowered it, whereas Na-citrate buffer improved it.

TABLE-US-00022 TABLE 17-1 Thermal stability results ID Formulation Tagg [° C.] Tunfold [° C.].sup.1 Tm [° C.] F1 20 mM histidine 65.2 66.8; 83.6 66.4; 70.3; 240 mM sucrose pH 86.3 6.0 F2 20 mM histidine 60.5 64.2; 80.4 63.8; 70.3; 240 mM sucrose pH 84.4 5.5 F3 20 mM histidine 64.5 66.6; 83.3 66.2; 70.7; 150 mM sucrose 50 86.2 mM arginine-HCl pH 6.0 F4 20 mM histidine 65.3 64.9; 79.2 64.6; 70.3; 150 mM NaCl pH 84.7 6.0 F5 20 mM citrate 67.5 68.0; 80.8 68.9; 72.5; 240 mM sucrose 85.9 pH 6.0

Colloidal Stability

[0347] Constant dissociation (kD) and osmotic second virial coefficient (A2) are both colloidal

stability indicators that measure interactions due to non-covalent forces between different molecules in solution. High values of kD and A2 indicate strong net repulsive interactions, whereas low values indicate net attractive forces. Whereas it is possible to differentiate between net attractive and net repulsive forces by the sign for A2, this is not possible for kD Formulations having an average good colloidal stability have an A2 value above 1.Math.10.sup.-4 mol.Math.ml.Math.g.sup.2.

[0348] Negative dissociation constants have been measured in all tested conditions, which correlates to weak negative or neutral osmotic second virial coefficient values and reflects a propensity for weak attractive protein-protein interactions.

[0349] The kD values can be ranked as follows: F2>F4>F3>F5>F1.

[0350] Colloidal stability is therefore improved by reducing the pH value to 5.5.

[0351] Increasing the ionic strength with NaCl, arginine-HCl or Na-citrate improves colloidal stability as well.

TABLE-US-00023 TABLE 17-2 Colloidal stability results														
kD A2 ID Formulation [ml/g] [mol .Math. mL .Math. g.sup.2]														
F1	20 mM histidine	-18.2	-6.9	10.sup.-5	240 mM sucrose	pH 6.0	F2							
20 mM histidine	-2.47	2.2	10.sup.-5	240 mM sucrose	pH 5.5	F3	20 mM histidine	-10.8	-2.6	10.sup.-5	150 mM sucrose	50 mM arginine-HCl	pH 6.0	F4
20 mM histidine	-5.03	7.3	10.sup.-6	150 mM NaCl	pH 6.0	F5	20 mM citrate	-11.6	-3.1	10.sup.-5	240 mM sucrose	pH 6.0		

[0352] Thus, the results of the determination of colloidal and thermal stability point in different directions when it comes to which pH to choose for the final formulation. While a pH of 6.0 would provide an improved thermal stability, the colloidal stability might deteriorate at this pH compared to pH 5.5.

[0353] This could potentially have a negative impact on the stability of the final formulation unless the negative impact on either of the two (thermal and colloidal stability, respectively) by choosing one over the other pH can be outweighed by carefully choosing other components of the formulation.

Early Stage Formulation Development

[0354] Once more in view of the results obtained with the experiments carried out so far, it was decided to still continue with liquid and lyophilized formulations.

[0355] In this further formulation set-up, the inventors came finally up with three formulations (two liquids and one lyophilized) which were to be tested under specific stress conditions. Target concentration for the antibody was set to 25 mg/ml.

[0356] The antibody to be stabilized in the present is CTL-002, as defined elsewhere herein.

TABLE-US-00024 TABLE 18-1 Compositions of formulations under evaluation

CTL-	25	20 mM	pH 150 mM	50 mM L-	0.02% w/v	2.4	Liquid	002_S6_FOR_F1	mg/mL	His/HCl	5.5	Sucrose	Arginine-	PS20	mL	HCl	CTL-	25	20 mM	pH 240 mM	—	0.02% w/v	2.4	Liquid	002_S6_FOR_F2	mg/mL	Na-citrate	6.4	Sucrose	PS80	mL	CTL-	25	20 mM	pH 150 mM	50 mM L-	0.02% w/v	2.4	Lyophilisate	002_S6_FOR_F3	mg/mL	His/HCl	5.5	Sucrose	Arginine	PS20	mL	HCl	CTL-	25	20 mM	pH 240 mM	—	0.02% w/v	2.4	Liquid	002_S6_FOR_F4	mg/mL	Na-Citrate	6.0	Sucrose	PS80	mL
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TABLE-US-00025 TABLE 18-2 Vial distribution and testing plan per formulation

—65° C.	25° C.	± 2°	40° C.	± 2°	or 5° C.	± C./60% RH	± C./75% RH	± Initial	below 3° C.	5% RH	5% RH	Liquid	Initial/T0	1	—	—	—	Freeze thaw:	—	1	—	—	—	from 25° C.	to -65° C.,	5 cycles	5 days	—	—	1	1	—	shaking stress.sup.@	4 weeks	—	—	—	1	1	8 weeks	—	—	1	1	1	Reserve	—	1	2	2	2	Vials/Temp	1	2	4	5	4	Target vials	16	Lyophilize	Initial/T0	3	—	—	—	—	4 weeks	—	—	—	—	1	8 weeks	—	—	1	1	2	Reserve	—	1	3	2	3	Vials/Temp	3	1	4	3	6	Target vials	17	.sup.@shaking stress	performed at ambient and cool temperatures as described under Methods section
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TABLE-US-00026 TABLE 18-3 Analytical testing plan per formulation

1	W	F/T.sup.\$	4	W	8	W	Incoming	Initial	Shaking.sup.\$	5	5°	25°	40°	5°	25°	40°	Analytics	DS	5° C.	Cool	Ambient	Cycles	C.	C.	C.	C.	C.	C.	Colour	—	x	x	x	x	—	x	x	x	x	x	Clarity/	—	x	x	x	x	—	x	x	x	x	x	opalescence	Visible
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— x x x x — x x x x x Particles Subvisible — x x x x — x x x x x particles (HIAC) pH x x x x x —
 x x x x x Osmolality — x — — — — — — — — — Protein — x x x x — x x x x x content Purity
 by Size x x x x x — x x x x x Exclusion- HPLC Purity by RP- x x x x x — x x x x x HPLC Caliper/
 x x x x x — x x x x x LabChip Non- Reduced and Reduced Purity by iCE x x x x x — x x — x x
 Surfactant — x x x x — x x x x x content Lyophilisate — x — — — — — x x x Reconstitution
 time.sup.1 Lyophilisate — x — — — — — x x x appearance and colour.sup.2 Lyophilisate — x
 — — — — — — — x x x Residual moisture content .sup.1only for lyophilized formulation
 .sup.2only for lyophilized formulation

Materials

[0357] Bulk purified drug substance (BPDS) of CTL-002 was supplied at a concentration of approximately 25 g/L

TABLE-US-00027 TABLE 19-1 Excipients and primary packaging materials

Material name (supplier)	Supplier name	Material Nr (supplier)
L-Histidine, USP, Multi-compendial	SAFC	H3911
L-Histidine, Monohydrochloride, FCC, SAFC	H4036	Multi-compendial
Sucrose, NF, Multi-Compendial, High Pfanstiehl	S-124-2-MC	Purity (Low Endotoxin)
Trisodium citrate dehydrate, Merck	1.37042.1000	MultiCompendial
Citric acid monohydrate, Merck	1.00243.1000	MultiCompendial
L-Arginine Hydrochloride	Sigma Aldrich	A5131
Polysorbate 20, N.F., Multi-J.T. Baker	4116-04	Compendial
TWEEN 20 HP-LQ-(MH) Polysorbate 80, NF, Multi-J.T. Baker	4117-04	compendial
(polyoxyethylene (80) sorbitan monooleate) CRILLET 4HP	Hydrochloric acid, NF, Multi-	Sigma Aldrich 320331
compendial Sodium Hydroxide 1N	Fluka 71686.00	6R
Schott Vials SL/NBE Tubular SCHOTT France	AC1063	Fiolax ® Clear Type I Pharmaceutical Systems or Adelphi Healthcare Packaging
20 mm Flurotec ® Injection stopper West pharmaceutical service	7001-8022 or Adelphi Healthcare or INJ20TB3WRS	20 mm Flurotec ® Freeze dry stopper Packaging
7001-9820 or FD20TT3WRS	20 mm True Edge ® Flip-off Seal (No West pharmaceutical service	AC1072 text)
Royal Blue		

TABLE-US-00028 TABLE 19-2 Main consumables

Supplier	Material Nr	Material name
Protein concentrator 30,000 MWCO, Pierce	88536	PES membrane Filter Stericup 0.22 µm, PVDF Millipore
SCGVU05RE/ membrane ScGVU02RE		Legend: PVDF Polyvinylidene fluoride, PES Polyethersulfone

TABLE-US-00029 TABLE 19-3 Main equipment

Equipment name	Model	Manufacturer
Centrifuge 5810R Eppendorf	Freeze dryer	LyoStar 3 FTS system
SP Scientific Shaker/s	KS 15B	Control Edmund Buhler
−65° C. and below Ultralow Dual, Panasonic freezer	MDFU700VX-PE	Stability 5° C. ± 3° C. n/a cold room
A7-47 Stability cabinet	KBF 720 Binder	25° C. ± 2° C./60% RH ± 5% RH
Stability cabinet	KBF 720 Binder	40° C. ± 2° C./75% RH ± 5% RH

[0358] Samples of all formulations were labelled and stored at 5±3° C. till distribution for the stability studies.

[0359] A testing sample per liquid formulation was subjected in horizontal position to shake stress during approximately 5 days at room temperature and cool temperature conditions in a reciprocating (horizontal) shaker at a target speed of 200 rpm.

[0360] A testing sample per liquid formulation was subjected in vertical position to five freeze/thaw cycles from −65° C. or below to room temperature.

[0361] Lyophilized formulation vials were reconstituted using 2.3 mL of purified water. Volume for reconstitution was calculated under consideration of volume displacement by solids. Upon reconstitution, vials were gently moved to assure completion of reconstitution and were used for further analysis.

Results

Formulations after Compounding

[0362] The pH, protein concentration, and osmolality of the compounded solutions for the liquid (F1, F2 and F4) and lyo (F3) formulations after compounding and filtration were determined.

Results are shown in Table 20. For completion, the protein concentration determined by UV

spectrophotometer (A280) at the initial timepoint during the short term stability study is also included. [0363] All results are close to the target values for pH [0364] The compounded solutions were colourless (BY7), and practically free of visible particles after compounding and filtration. TABLE-US-00030 TABLE 20 Results of characterization of formulations after compounding

Formulations	CTL-002	S6	FOR	sub.—	Test	F1	F2	F3	F4	pH	.sup.a	5.5	6.4	5.6	6.0	Osmolality by freezing	328	337	306	367	point [mOsm/Kg]	.sup.a	Protein concentration	.sup.a	[mg/mL]	25.5	25.5	26.2	23.1	.sup.a	values taken over from TO results
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Freeze-Thaw and Shaking Studies

[0365] All formulations, except F3, subjected to 5 freeze-thaw cycles (-65°C . to RT) or shaking stress at ambient temperature and at cool temperature did not show any relevant changes in any of the analytical methods compared to the initial non-stressed samples, indicating that the formulations effectively stabilized the CTL-002 molecule against both freeze-thaw and shaking stress.

[0366] For all liquid formulations no major differences were observed in aggregation and fragmentation by SE-HPLC. Also, no major chemical degradation has been observed by iCE and overall visible and subvisible particle counts were low on shaking and F/T stress. Data suggested that both surfactants, Polysorbate 20 and 80 protect formulations against the shaking, freezing and thawing stresses.

Short-Term Stability Studies

[0367] Overall, pH and protein concentration remained stable in all four tested formulations over the short-term stability testing up to 8 weeks. The stability testing revealed initial low subvisible particle counts for all formulations. The lyophilizate formulation F3 demonstrated a tendency for higher level of subvisible particles as compared to the liquid formulations F1, F2 and F4, which is inherent to the lyo cake reconstitution. No significant change in subvisible particle counts was detected in all tested formulations after 8-week storage at any of the tested storage conditions. Moreover, all samples were practically free from visible particles at T0 as assessed during the visual inspection using a black and white background. Only formulation F4 demonstrated an increase of the visible particle counts where few particles, white fibres, were observed after 8-week storage at 25°C . This increase was however not confirmed at 40°C . Overall, the level of visible particles did not change over the short-term stability studies.

[0368] The lyophilizate formulation F3 demonstrated a high stability. No change in purity by SE-HPLC, icIEF, RP-HPLC and CE-SDS could be observed over the 8-week stability stress under all tested storage conditions. However, the results for the liquid formulations, but in particular formulation F1, showed that it was possible to stabilize the antibody suitably even in a liquid formulation.

[0369] The CTL-002 molecule had a low aggregation and fragmentation tendency in all three tested liquid formulations when stressed at 40°C . The loss in monomer by aggregation as well as by fragmentation was more pronounced in formulation F1 than in formulations F2 and F4. After an 8-week storage at 40°C . 1.2% aggregates and 0.3% degradation products were measured in F1 (pH 5.5), whereas formulations F2 (pH 6.4) and F4 (pH 6.0) contained around 1.0-0.9% aggregates and 0.1-0.2% degradation products. The formation of high molecular weight species and low molecular weight species is therefore considered to be pH dependent.

[0370] The low solution turbidity, the low level of subvisible particles and the absence of visible particle measured in all tested formulations over the entire stability study underlined the low aggregation tendency. Moreover, no trend for aggregation nor fragmentation could be detected by chip-based CE-SDS, where no changes under normal and reduced conditions could be observed.

[0371] The chemical purity of CTL-002 was modified by thermal stress at 40°C . and to a lesser extent at 25°C . as measured by icIEF. A loss in main peak purity was more pronounced in formulations F2 and F4 than in formulation F1 and was mainly attributed to the formation of acidic species. Only formulation F1 at pH 5.5 showed in addition a significant uptake of basic species.

[0372] The RP-HPLC analysis under non-reduced conditions showed in all liquid formulations a loss in main peak followed by a post-peak species increase at 40° C., when no changes could be noticed by using RP-HPLC with reduced conditions. The RP-HPLC changes were as in icIEF less pronounced in formulation F1 than in formulations F2 and F4. Formulation F1 having a lower pH 5.5 demonstrated altogether a higher chemical stability than formulations F2 and F4.

[0373] No significant changes in polysorbate content could be observed in all formulations over the entire short-term stability study. Both surfactants, polysorbate 20 and polysorbate 80 are suitable for CTL-002 formulation.

Conclusions:

[0374] The lyo formulation F3 was stable during the entire short-term stability study. [0375] All liquid formulations were stable against freeze/thaw and shaking stress as well as upon storage for 8 weeks at 5° C. Formulations F2 and F4 demonstrated similar stability profiles overall the tested stressed conditions. [0376] The lowest pH formulation F1 was more prone to aggregation and fragmentation under accelerated storage conditions at 40° C. [0377] The lowest pH formulation, F1, demonstrated a higher chemical stability than formulations F2 and F4 at 40° C. and to a lesser extent at 25° C. by icIEF, where mainly the acidic species propensity increased under stressed conditions, as well as at 40° C. by RPHPLC, where post peak species increased.

[0378] Based on the results of all experimental data taken together as performed for this formulation project up to this point, the following liquid formulation was provided for CTL-002:

[0379] 25 mg/mL CTL-002, 20 mM Histidine/Histidine HCl, 150 mM sucrose, 50 mM Arginine-HCl, 0.02% w/v Polysorbate 20, at pH 5.5

[0380] At this stage, it was very difficult to make that decision, as the data (see above) seemed to be somewhat contradictory for chemical stability on the one hand and physical stability on the other hand. However, the inventors—taking together all data obtained up to this point—found that chemical stability would be particularly important in the context of the stabilization efforts for this antibody.

Long Term Data

[0381] With the above described formulation for the IgG4 antibody CTL-002, two long term stability studies were performed.

[0382] Stability samples for the product CTL-002 (at 250 mg/10 mL) stored at long term storage conditions 5° C.±3° C. inverted and upright were tested.

[0383] After eighteen months at long-term storage conditions 5° C.±3° C. inverted and upright storage, the product CTL-002 shows no degradation by the stability indicating methods: [0384] SE-HPLC (main peak, fragments, aggregates) and [0385] CE-DSD (reduced sum LC+HC, non-reduced intact IgG).

[0386] Only a slight shift from main peak towards acidic species is indicated by icIEF results. A decrease in polysorbate content can also be detected.

[0387] As can be seen from the results, the antibody was stabilized to a high extent not only with regard to the chemical stability parameters determined during this study, but—very surprisingly—also with regard to its aggregation properties.

Sequences

TABLE-US-00031 SEQ ID No: 1 (Heavy Chain CDR1 Region Peptide Sequence of monoclonal anti-human GDF-15 antibody): GFSLSTSGMG SEQ ID No: 2 (Heavy Chain CDR2 Region Peptide Sequence of monoclonal anti-human GDF-15 antibody): IYWDDDK SEQ ID No: 3 (Heavy Chain CDR3 Region Peptide Sequence of monoclonal anti-human GDF-15 antibody): ARSSYGAMDY SEQ ID No: 4 (Light Chain CDR1 Region Peptide Sequence of monoclonal anti-human GDF-15 antibody): QNVGTN Light Chain CDR2 Region Peptide Sequence of monoclonal anti-human GDF-15 antibody: SAS SEQ ID No: 5 (Light Chain CDR3 Region Peptide Sequence of monoclonal anti-human GDF-15 antibody):

QQYNNFPYT SEQ ID No: 6 (heavy chain variable domain of monoclonal anti-human GDF-15 antibody):

QITLKESGPTLVKPTQTLTLTCTFSGFSLSTSGMGVSWIRQPPGK
GLEWLAHIYWDDDKRYNPTLKSRLTITKDPSKNQVVLTMNMDPV

DTATYYCARSSYGAMDYWGQGLTVTVSSASTKGP SEQ ID No: 7 (light chain variable domain of monoclonal anti-human GDF-15 antibody):

DIVLTQSPSFLSASVGDRVTITCKASQNVGTNVAWFQQKPGKSPK
ALIYSASYRYSQVPDRFTGSGSGTEFTLTISLQPEDFAAYFCQQ

YNNFPYTFGGGKLEIKRT SEQ ID No: 8 (heavy chain of monoclonal anti-human GDF-15 antibody CTL-002 without the leader peptide sequence):

QITLKESGPTLVKPTQTLTLTCTFSGFSLSTSGMGVSWIRQPPGK
GLEWLAHIYWDDDKRYNPTLKSRLTITKDPSKNQVVLTMNMDPV
DTATYYCARSSYGAMDYWGQGLTVTVSSASTKGPSVFPLAPCSRS
TSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGL
YSLSSVVTVPSSSLGTQTYTCNVDHKPSNTKVDKRVESKYGPPCP
PCPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSDQEDPE
VQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVLHQDWLNGK
EYKCKVSNKGLPSSIEKTISKAKGQPREPQVYTLPPSQEEMTKNQ
VSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLY

SRLTVDKSRWQEGNVFSCSVMHEALHNHYTQKSLSLGL SEQ ID No: 9 (light chain of monoclonal anti-human GDF-15 antibody CTL-002 without the leader peptide sequence):

DIVLTQSPSFLSASVGDRVTITCKASQNVGTNVAWFQQKPGKSPK
ALIYSASYRYSQVPDRFTGSGSGTEFTLTISLQPEDFAAYFCQQ
YNNFPYTFGGGKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCL
LNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTYSLSSTLT

LSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC SEQ ID NO: 10 (heavy chain variable domain of anti-human GDF-15 antibody H1L5):

QITLKESGPTLVKPTQTLTLTCTFSGFSLSTSGMGVSWIRQPPGK
GLEWLAHIYWDDDKRYNPTLKSRLTITKDPSKNQVVLTMNMDPV
DTATYYCARSSYGAMDYWGQGLTVTVSSASTKGPSVFPLAPSSKS
TSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGL
YSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTH
TCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSF

FLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPG SEQ ID NO: 11 (light chain variable domain of anti-human GDF-15 antibody H1L5):

DIVLTQSPSFLSASVGDRVTITCKASQNVGTNVAWFQQKPGKSPK
ALIYSASYRYSQVPDRFTGSGSGTEFTLTISLQPEDFAAYFCQQ
YNNFPYTFGGGKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCL
LNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTYSLSSTLT
LSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC

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K, Kim J T, Kim H W, et al. GDF15 deficiency exacerbates chronic alcohol- and carbon tetrachloride-induced liver injury. *Sci Rep*. 2017; 7(1):1-13. doi:10.1038/s41598-017-17574-w [0391] Eisenhauer E A, Therasse P, Bogaerts J, et al. New Response Evaluation Criteria in Solid Tumours: Revised RECIST Guideline (Version 1.1). *Eur J Cancer*. 2009 January; 45(2):228-47. [0392] Emmerson, Paul J., Feng Wang, Yong Du, Qian Liu, Richard T. Pickard, Malgorzata D. Gonciarz, Tamer Coskun, et al. 2017. "The Metabolic Effects of GDF15 Are Mediated by the Orphan Receptor GFRAL." *Nature Medicine* 23 (10): 1215-19. <https://doi.org/10.1038/nm.4393>. [0393] Hodi F S, Ballinger M, Lyons B, et al. Immune-Modified Response Evaluation Criteria In Solid Tumors (imRECIST): Refining Guidelines to Assess the Clinical Benefit of Cancer Immunotherapy. *J Clin Oncol*. 2018 Mar. 20; 36(9):850-858. [0394] Johnen H, Lin S, Kuffner T, et al. Tumor-induced anorexia and weight loss are mediated by the TGF- β superfamily cytokine MIC-1. *Nat Med*. 2007; 13(11):1333-1340. doi:10.1038/nm1677 [0395] Kempf, Tibor, Alexander Zarbock, Christian Wiedera, Stefan Butz, Anika Stadtmann, Jan Rossaint, Matteo Bolomini-Vittori, et al. 2011. "GDF-15 Is an Inhibitor of Leukocyte Integrin Activation Required for Survival after Myocardial Infarction in Mice." *Nature Medicine* 17 (5): 581-88. <https://doi.org/10.1038/nm.2354>. [0396] Selby, Mark J., John J. Engelhardt, Robert J. Johnston, Li-Sheng Lu, Minhua Han, Kent Thudium, Dapeng Yao, et al. 2016. "Preclinical Development of Ipilimumab and Nivolumab Combination Immunotherapy: Mouse Tumor Models, In Vitro Functional Studies, and Cynomolgus Macaque Toxicology." Edited by Aamir Ahmad. *PLOS ONE* 11 (9): e0161779. <https://doi.org/10.1371/journal.pone.0161779>. [0397] Tong S, Marjono B, Brown D A, et al. Serum concentrations of macrophage inhibitory cytokine 1 (MIC 1) as a predictor of miscarriage. *Lancet*. 2004; 363(9403):129-130. doi:10.1016/S0140-6736(03)15265-8 [0398] Tsai, Vicky W. W., Yasmin Husaini, Amanda Sainsbury, David A. Brown, and Samuel N. Breit. 2018. "The MIC-1/GDF15-GFRAL Pathway in Energy Homeostasis: Implications for Obesity, Cachexia, and Other Associated Diseases." *Cell Metabolism* 28 (3): 353-68. <https://doi.org/10.1016/j.cmet.2018.07.018>. [0399] van den Boom, Jasper G., and Gunther Hartmann. 2013. "Turning Tumors into Vaccines: Co-Opting the Innate Immune System." *Immunity* 39 (1): 27-37. <https://doi.org/10.1016/j.immuni.2013.07.011>. [0400] Vaupel, Peter. 2004. "Tumor Microenvironmental Physiology and Its Implications for Radiation Oncology." *Seminars in Radiation Oncology* 14 (3): 198-206. <https://doi.org/10.1016/j.semradonc.2004.04.008>. [0401] Wang M, Yao L C, Cheng M, Cai D, Martinek J, Pan C X, et al. Humanized mice in studying efficacy and mechanisms of PD-1-targeted cancer immunotherapy. *FASEB J*. 2018; 32(3):1537-49. [0402] Welsh, John B., Lisa M. Sapinoso, Suzanne G. Kern, David A. Brown, Tao Liu, Asne R. Bauskin, Robyn L. Ward, et al. 2003. "Large-Scale Delineation of Secreted Protein Biomarkers Overexpressed in Cancer Tissue and Serum." *Proceedings of the National Academy of Sciences* 100 (6): 3410-15. <https://doi.org/10.1073/PNAS.0530278100>. [0403] Wischhusen J, Melero I, and Fridman W. GDF-15: From Biomarker to Novel Targetable Immune Checkpoint. *Front. Immunol*. Accepted 23 Apr. 2020. doi: 10.3389/fimmu.2020.00951. [0404] Wollert K C, Kempf T, Giannitsis E, et al. An Automated Assay for Growth Differentiation Factor 15. *J Appl Lab Med An AACC Publ*. 2018; 1(5):510-521. doi:10.1373/jalm.2016.022376

INDUSTRIAL APPLICABILITY

[0405] The anti-GDF-15 antibody may be used in methods for the treatment of cancer in human patients can be industrially manufactured and sold as products for the itemed methods and uses, in accordance with known standards for the manufacture of pharmaceutical products. Accordingly, the present invention is industrially applicable.

Claims

1. An anti-GDF-15 antibody, wherein the antibody comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 8 and a light chain comprising the amino acid sequence of SEQ ID

NO:9.

2. The anti-GDF-15 antibody of claim 1, wherein the antibody is produced by expression in CHO cells.
3. A formulation comprising 10-50 mg/ml of the antibody of claim 1.
4. The formulation of claim 3, further comprising histidine/histidine HCl, sucrose, arginine-HCl, and polysorbate, at a pH between 5 and 6.
5. The formulation of claim 3, further comprising 10-50 mg/ml histidine/histidine HCl, 100-200 mM sucrose, 20-80 mM arginine-HCl, and 0.01 to 0.05% w/v polysorbate 20 or polysorbate 80, at a pH between 5.0 and 6.0.
6. The formulation of claim 5, wherein the pH is between 5.3 and 5.7.
7. The formulation of claim 3, comprising 25 mg/ml of the antibody, 20 mM histidine/histidine HCl, 150 mM sucrose, 50 mM arginine-HCl, and 0.02% w/v polysorbate 20, at pH 5.5.
8. The formulation of claim 3, consisting of 25 mg/ml of the antibody, 20 mM histidine/histidine HCl, 150 mM sucrose, 50 mM arginine-HCl, and 0.02% w/v polysorbate 20, at pH 5.5.
9. A method of treating cancer and/or cancer cachexia in a human patient, the method comprising administering to the patient an anti-GDF-15 antibody, wherein the antibody is administered at a dose of 0.3, 1.0, 3.0, 10.0 or 20.0 mg/kg, and at a dosage regimen of at least one administration cycle, wherein the cycle is a period of two weeks, three weeks, or four weeks, and wherein the dose is administered at least once in each cycle.
10. The method of claim 9, wherein the anti-GDF-15 antibody is administered at a dose of 3, 10 or 20 mg/kg.
11. The method of claim 9, wherein the anti-GDF-15 antibody is administered at a dose of 10 mg/kg.
12. The method of claim 9, wherein the antibody does not induce antibody-dependent cell-mediated cytotoxicity (ADCC) and/or does not induce Complement Dependent Cytotoxicity (CDC).
13. The method of claim 9, wherein the concentration of GDF-15 in the serum or plasma of the patient is below 10 ng/mL at the end of an administration cycle.
14. The method of claim 13, wherein the concentration of GDF-15 in the serum of the patient is below 2 ng/mL at the end of an administration cycle.
15. The method of claim 13, wherein the concentration of GDF-15 in the serum of the patient is below 0.5 ng/mL at the end of an administration cycle.
16. The method of claim 9, wherein the cycle is a period of three weeks and the dosage regimen consists of up to 34 administration cycles.
17. The method of claim 16, wherein the dosage regimen consists of 17 to 34 administration cycles.
18. The method of claim 9, wherein the cycle is a period of four weeks and wherein the dosage regimen consists of up to 24 administration cycles.
19. The method of claim 18, wherein the dosage regimen consists of 12 to 24 administration cycles.
20. The method of claim 9, wherein the dosage regimen consists of up to 52 administration cycles.
21. The method of claim 20, wherein the dosage regimen consists of 26 to 52 administration cycles.
22. The method of claim 9, wherein the anti-GDF-15 antibody is administered in combination with a checkpoint inhibitor, optionally wherein the checkpoint inhibitor is selected from the group consisting of an anti-PD-1 antibody, an anti-PD-L1 antibody, and an anti-CD40 antibody.
23. The method of claim 22, wherein the checkpoint inhibitor is administered at the same dosage regimen as the anti-GDF-15 antibody.
24. The method of claim 22, wherein the checkpoint inhibitor is administered prior to the administration of anti-GDF-15 antibody.
25. The method of claim 24, wherein the checkpoint inhibitor is administered within 120 minutes prior to the administration of the anti-GDF-15 antibody.
26. The method of claim 25, wherein the checkpoint inhibitor is administered within 30 minutes

prior to the administration of the anti-GDF-15 antibody.

27. The method of claim 9, wherein the antibody comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 8 and a light chain comprising the amino acid sequence of SEQ ID NO:9.

28. The method of claim 9, wherein the cancer is selected from the group consisting of brain cancers including glioma, cancers of the nervous system, melanoma, lung cancer, head and neck cancer, urothelial cancer, liver cancer, endometrial cancer, cervical cancer, gastric cancer, renal cell carcinoma, Ewing's sarcoma, non-small cell lung cancer, small cell lung cancer, lip and oral cavity cancer, hepatic carcinoma, leukemia, Hodgkin lymphoma, Non-Hodgkin lymphoma, bladder cancer, cervix uteri cancer, corpus uteri cancer, testis cancer, thyroid cancer, kidney cancer, gallbladder cancer, multiple myeloma, nasopharynx cancer, larynx cancer, pharynx cancer, oesophagus cancer, gastrointestinal tumor, stomach cancer, colorectal cancer, pancreatic cancer, prostate cancer, ovarian cancer, breast carcinoma, and carcinoma of unknown primary.

29. The method of claim 9, wherein the dose of the anti-GDF-15 antibody is administered intravenously.

30. A formulation comprising an anti-GDF-15 antibody, wherein: (a) the formulation comprises 10-50 mg/ml of the antibody; (b) the formulation comprises histidine/histidine HCl, sucrose, arginine-HCl, and polysorbate, at a pH between 5 and 6; (c) the formulation comprises 10-50 mg/ml histidine/histidine HCl, 100-200 mM sucrose, 20-80 mM arginine-HCl, and 0.01 to 0.05% w/v polysorbate 20 or polysorbate 80, at a pH between 5.0 and 6.0; (d) the formulation comprises 10-50 mg/ml histidine/histidine HCl, 100-200 mM sucrose, 20-80 mM arginine-HCl, and 0.01 to 0.05% w/v polysorbate 20 or polysorbate 80, at a pH between 5.0 and 6.0; (e) the formulation comprises 25 mg/ml of the antibody, 20 mM histidine/histidine HCl, 150 mM sucrose, 50 mM arginine-HCl, and 0.02% w/v polysorbate 20, at pH 5.5; and/or (f) the formulation consists of 25 mg/ml of the antibody, 20 mM histidine/histidine HCl, 150 mM sucrose, 50 mM arginine-HCl, and 0.02% w/v polysorbate 20, at pH 5.5.
